



AI and human autonomy: a literature review

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Abstract

The interaction with AI technologies has opened a debate about its impact on a core value and concept, namely human autonomy. However, this debate is largely heterogeneous and unstructured. This is the first review dedicated to AI ethics' scholarly literature concerning the impact on human autonomy. The aim is to map and structure this debate by surveying its key concepts, topics, and gaps. The first part clarifies how human autonomy is understood in the reviewed literature by identifying its multiple conceptions. This shows how different accounts of autonomy underpin different analyses of the impact of AI technologies. The second part maps and explains the ways AI technologies affect the concept and exercise of human autonomy. Two main sets of concerns emerge: first, AI mediation of human activities influences autonomous deliberation, decision-making, and behaviour; second, AI automation of human activities affects the conditions for autonomous choice and action. This categorisation shows how the impact of AI technologies depends on the system, the context, and the conception of autonomy. This literature review develops a common framework providing a comprehensive understanding of the main concepts and issues at stake, as well as indicating future research directions.

Keywords Human autonomy · AI technologies · Literature review · AI ethics

1 Introduction

Within and beyond the Western philosophical tradition, human autonomy is commonly considered a core value and right. As proof of its relevance, autonomy is central to individual and social issues as varied as responsibility, well-being, and democracy. In recent years, the importance of human autonomy has further increased due to the advancement of AI technologies that mediate or automate human decisions and actions. In this case, human autonomy is evoked in debates ranging from manipulation and surveillance to personalised assistance and moral enhancement. Thus, the interaction with AI technologies has crucial implications for human autonomy: it entails opportunities and threats for its exercise, as well as challenges for its conceptualisation and operationalisation.

For all these reasons, a growing amount of scholarly and policy literature in AI ethics addresses the relation between human autonomy and AI technologies. However, both these scholarly and policy debates are largely heterogeneous and unstructured: there is no shared definition of human autonomy but various competing conceptions, nor is there a coordinated assessment of the impact of AI technologies but multiple alternative analyses (cf. [1–3]). Moreover, the debate is divided between stressing, on the one hand, the threats and challenges, on the other hand, the opportunities and benefits of AI technologies for human autonomy. While a plurality of views allows for a deeper and richer discussion, the lack of structure undermines a comprehensive understanding and a coordinated assessment of how AI technologies affect human autonomy.

The development of a heterogeneous and complex debate motivates an overview of the existing literature to clarify the state of the art and to guide future research. This paper contributes to this goal through the first review entirely dedicated to map, synthesise, and structure AI ethics' academic literature on human autonomy. More precisely, this review offers the following contributions: mapping and synthesising the literature through an overview of the main topics and concepts, integrating the heterogeneous literature in a

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common framework to support future research, and identifying gaps or open questions for new lines of inquiry.¹

After illustrating the methodology (Sect. 2), the review is structured in two main parts. First, it clarifies how human autonomy is understood in AI ethics by surveying various conceptions employed in the literature (Sect. 3). This clarification shows how different accounts of autonomy underpin different analyses and assessments of the impact of AI technologies. Second, it maps the ways AI technologies affect human autonomy by categorising and explaining the main issues discussed in the literature (Sect. 4). This categorisation shows how the impact of AI technologies varies depending on the system, the context, and the conception of autonomy. The results of both Sections are first described qualitatively (Findings subsections) and then critically analysed (Discussion subsection). In both Sections, the reviewed literature is compared with previous philosophical debates on autonomy when helpful to clarify the current literature. Section 5 concludes with final comments on the main results, contributions, and open questions. This literature review is complemented with three Appendices that include a detailed methodology, the database of reviewed literature, three tables of results, and a glossary of the main terms employed in the review. For brevity, the expression “AI technologies” is henceforth replaced with the term “AI”.²

¹ Within the literature on AI, this review doesn’t deal with the debate on *machine autonomy* (i.e., the philosophical and ethical issues concerning the development of artificial systems that perform tasks with less or no human intervention and oversight, (see e.g., [4, 5]), unless it has implications for human autonomy (see Sect. 4.2.9). For an overview of autonomy in AI research see e.g. [6]. This review doesn’t focus on AI ethics’ policy literature on the principle of autonomy (for a critical analysis of such literature see e.g., [1, 7]). Rather it focuses on the growing academic, mostly philosophical, literature concerning the relation between human autonomy and AI systems. A synthetic overview of this academic debate is included in some of the reviewed papers as a background for their analyses (e.g., [3, 8]). However, this is the first review entirely and exclusively dedicated to this literature. Therefore, this work complements the other overviews in the following ways. First, by covering relevant literature until December 2024 (including papers that were first published online in 2024, and were then published in an issue in 2025; see Table 1 and the References for details). Second, by surveying, mapping, and classifying both the conceptions of human autonomy and the analyses of AI technologies’ impacts. Third, by developing a framework that integrates various contributions in the reviewed literature without eliminating its heterogeneity.

² There isn’t a standard definition of AI, but the term has been used in different ways and to describe various technologies (see also [9, 10]). Broadly, AI technologies are computer systems that perform tasks resembling intelligent behaviour, with some degrees of independence, and in a goal-directed way. Many AI technologies are based on machine learning, a method through which algorithms learn from data – via supervised, unsupervised, or reinforcement learning – to make inferences on new data without being explicitly programmed. In this review, the term “AI” or “AI technologies” is used in a broad

2 Methodology

This literature review follows the methodological approach of critical or narrative philosophical reviews (see e.g., [11–14] for examples). The aim is to synthesise, analyse, and evaluate the reviewed literature to understand and advance the ongoing debate, emphasising the arguments, theories, and concepts presented. The identification of relevant material started with literature already known by the author. This was complemented by a more structured search and selection through databases commonly used in philosophy (PhilPapers, Google Scholar). This process was reiterated during the writing phase to ensure that no recent important publication was missing. The search focused on the philosophical and ethical literature discussing the impact of AI on human autonomy. The selection of key themes was conducted through an in-depth analysis of the texts, aimed at identifying key concepts and issues, common patterns and trends, as well as gaps and open questions. This process started by using pre-existing philosophical conceptions and issues of autonomy as the background guiding the identification of similar themes in the reviewed literature. This approach remained flexible enough to leave room for variations or new themes emerging from the literature for a more comprehensive overview. The review followed a qualitative description of results, combined with a critical analysis. The methodology is illustrated in details in the Appendix 1.

3 The concept of human autonomy in AI ethics

3.1 Background

Autonomy is a core value and concept in the Western philosophical tradition and beyond. However, it is also complex and multifaceted, subject to different interpretations and accounts (see [15, 16] for an overview). While it is broadly understood as self-governance, its meaning has changed throughout the history of moral and political philosophy: from the self-governing city in Ancient Greece, to the modern conception of moral autonomy by Kant, to contemporary accounts of personal autonomy and the principle of patient autonomy. Traditional categories of autonomy include moral (i.e., the capacity to give oneself a moral law), personal (i.e., the capacity to live a life according to one’s values and goals), political (i.e., the capacity of self-determination in the public sphere), and epistemic (i.e., the capacity to think for oneself). Contemporary debates have

way that mostly concerns, without being limited to, machine learning systems. A wide range of AI applications are included, from recommender systems, to health apps, to social robots, to chatbots.

mostly focused on personal autonomy, producing a variety of accounts distinguished along two main lines (cf. [15]): First, *internalism* only includes internal individual elements (e.g., psychological capacities, desires, or beliefs), while *externalism* also considers external elements (e.g., reasons, relations, social context). Second, *proceduralism* only considers the right process of forming motives (e.g., reflective endorsement, identification, non-alienation), while *substantivism* also considers the right content of motives (e.g., non-oppressive). Different contemporary accounts thus identify various internal and external conditions of autonomy, including *competency* (range of rational skills and self-attitudes), *authenticity* (internal coherence or non-alienation with one's true motives, through reflective endorsement), *freedom* or independence, *opportunities* (range of available and viable options), as well as *social support*.³ This Section clarifies how the concept and principle of human autonomy is understood in AI ethics by surveying and classifying different conceptions (i.e., ways of understanding and specifying a concept) in the literature. A detailed illustration of the resulting framework is provided in Fig. 2 of Appendix 2.⁴

3.2 Findings

Overall, the reviewed literature doesn't provide a common definition of human autonomy, but presents a variety of alternative and partially unrelated notions and interpretations (similar to what is observed in AI ethics guidelines and to what has been the case in previous philosophical literature; see [1, 15]).

At the highest level, the reviewed literature broadly characterises human autonomy with the traditional notions of *self-governance*, *self-determination*, or *self-rule* (e.g. [3, 23, 24]). These broad characterisations are closely related, but can be interpreted and specified in different ways. Among the traditional categories of autonomy, about half of the reviewed literature concerns *personal* or *individual* autonomy (e.g., [3, 8, 23, 25–33]), while a smaller subset of about ten publications addresses issues for *moral* autonomy (e.g., [34–38]). Other categories are much rarer or less explicitly

addressed: for instance, a discussion of the political role of autonomy, and how it is affected by technologies, can be found in Rubel et al. [39]; the impact on epistemic autonomy is addressed by a paper on AI deepfake detectors [40], and indirectly by e.g., literature on the impact of recommender systems on critical thinking [25]. Additionally, *decisional* autonomy [41] figures either as a broader category or as dimension of other categories. As a component of both personal and moral autonomy (e.g., [8, 35, 42]), autonomous decision-making is the capacity to make competent independent decisions on the basis of one's motives, variously spelt out. As a separate category, decisional autonomy can be seen as a “meta-autonomy”: the “power to decide (whether to decide)”, expressing a balance between freedom of choice and (reversible) delegation to AI [43].

While not all the reviewed papers specify a theory or model of autonomy, the review identifies various accounts of personal and moral autonomy (i.e., theories and models explaining their definitions and features) underpinning the analysis of AI's impact. Amidst this variety, there are some patterns that allow for the classification of various accounts into groups sharing key features, despite remaining quite diverse from one another. The proposed classification is based on the similarity of these accounts with pre-existing theories and models, while leaving room for new combinations or new groups emerging from the reviewed literature. Even if the heterogeneity of the literature makes neat and exhaustive classification difficult, the proposed framework collects and connects various accounts without eliminating their differences.

Indeed, while some heterogeneity in the conception of autonomy is inevitable and enriching, the lack of a common framework undermines a comprehensive understanding of the debate. To cope with this issue, the following subsections organise various conceptions of human autonomy in the reviewed literature on the basis of the category (i.e., a group of conceptions pertaining to the same domain) and the respective accounts (i.e., theoretical frameworks that specify the definition and features of a category).⁵ A detailed overview presenting together the categories and the

³ Among the main contemporary accounts, there are: structuralist accounts that base autonomy on either a hierarchical [17, 18] or diachronic [19] relation between one's preferences and desires; self-authorship accounts [20] that conceive autonomy as a relation with one's life plans; relational accounts that consider autonomy as caused or constituted by social relations [21, 22].

⁴ Before and beyond AI ethics, autonomy has been discussed in a wide variety of debates, not only in moral philosophy, but also in cognitive science, artificial life, cybernetics, robotics, etc. Clearly, the understanding of autonomy largely varies across and within these debates. For instance, autonomy is often understood as behavioural self-sufficiency in engineering and robotics, and as self-organisation in artificial life (for an overview of these other conceptions and debates; see e.g. [6]. I thank the anonymous reviewer for the reference).

⁵ Since the literature on autonomy lacks a shared and standard terminology, the terms used here (e.g., conceptions, characterisations, categories, accounts, etc.) might differ from those encountered elsewhere. In the terminology adopted by the review, the concept of autonomy is the overarching notion. Different conceptions of autonomy are different ways of understanding and specifying what the concept of autonomy is about (see [44]). The review maps and unpacks these conceptions by spelling out the following elements: characterisations (i.e., basic general descriptions, such as self-governance and self-rule), categories (i.e., classes of conceptions, such as personal and moral), and accounts (i.e., theoretical frameworks spelling out definitions and features of the respective category). Other terms employed in the review (e.g., models, dimensions, components, etc.) are retrieved from the reviewed literature with their original meaning,

respective accounts, as well as related topics addressed, is provided in Table 3 of Appendix 2.

3.2.1 Philosophical conceptions of moral and ethical autonomy

As formulated by Kant, moral autonomy is the capacity of the will to give a moral or universal law to itself. The literature stresses different facets of moral autonomy, depending on different conceptions, contexts, and issues. The closest to Kant's original formulation is Chiodo [24, 45], who criticises the implications of technological automation (autonomous vehicles and self-tracking technologies) on human self-given rule (i.e., reflecting on and choosing in accordance to reasons, as well as being responsible and accountable). In the context of driving automation through Autonomous Vehicles, moral autonomy broadly concerns the capacity to make one's own decisions on morally relevant issues, such as risk allocation [8].⁶ In the debate on AI assistants for ethical decision-making, moral autonomy indicates the ability to form, reflect, decide, and act in accordance with one's moral values or ethical principles (e.g., [35, 46]). In relation to algorithmic recommendations, two contributions [37, 38] consider moral autonomy as the condition of being the authors of our decisions and actions, requiring the capacity to reflectively endorse the motives at the basis of our moral choices.

While formally similar to the latter, Pierosara [47] introduces a separate conception of autonomy, different from both moral and personal autonomy, only attributable to humans (more on this in Sect. 4.2.9): *ethical autonomy* is thus defined by Pierosara [47] as "the capacity to deliberate and act without constraints following an interpretation and endorsement of what we consider valuable". Ethical autonomy is further spelt out in terms of *narrative autonomy*, consisting in making meaning in an open, non-deterministic way.

3.2.2 Philosophical conceptions of personal autonomy

Although not always explicit, personal autonomy is understood by the literature in the traditional sense (i.e., making and enacting choices aligned with one's deep motives) and approached in the descriptive sense explained by Bonicalzi et al. [25]: *Descriptive Autonomy* is a range of features required to be (considered) autonomous, while *Normative Autonomy* is the right to be treated as autonomous agents. Despite referring to the same category of autonomy, the

literature disagrees about its interpretation: together with rational deliberation and decision-making, there is an increasing recognition of external factors (especially, social relations) underpinning the capacity to pursue one's goals and life plans. The following details the main accounts of personal autonomy identified in the literature.

Among traditional accounts of autonomy, Kant's is employed by Aylsworth and Castro [48] to conceptualise personal autonomy. However, rather than referring to the most well-known notion of self-rule, the authors adopt Kant's concept of "humanity": "the rational agent's capacity to set and pursue her own ends" [48]. According to them, the Kantian duty to respect autonomy/humanity justifies avoiding specific cases of AI outsourcing (see Sect. 4.2.6). A further different Kantian-Piagetian account of autonomy is developed by Rini [49] to show that human self-control (i.e., capacity of deliberation and accountability to reason) is undermined by chatbots that favour our authoritarian tendencies over them.

However, the majority of the conceptions of personal autonomy in the literature are influenced by *contemporary accounts*.⁷ Internalist and proceduralist accounts (e.g., Frankfurt's hierarchical account and Christman's historical account) more or less directly underpin some of the analyses and explanations addressing AI's influence on internal elements (e.g., preferences) and conditions of autonomy, especially of autonomous decision-making (see e.g., [3, 25]). Externalist accounts such as Raz's self-authorship [20] or feminist socio-relational accounts (e.g., [21, 22]) are employed by the literature to highlight that AI also affects external conditions of autonomy (see e.g., [3, 26, 39, 50]). Among them, five papers [8, 26, 27, 51, 52] refer to Raz's model of autonomy as *self-authorship* to assess how cases of AI automation (e.g., artificial assistants, automated route recommendations) affect three core conditions of personal autonomy: *rationality* (i.e., understanding the reasons for choice), *optionality* (i.e., adequate range of available and viable alternatives), *independence* (i.e., freedom from coercion and manipulation) [8, 26].

While still underexplored, the literature is turning from traditional individualistic accounts (e.g., Kantian, internalist) to relational accounts of autonomy, especially from the Western feminist literature (by e.g., Mackenzie and Stoljar or Oshana [21, 22]) and also from non-Western philosophical traditions (e.g., Ubuntu). Notwithstanding the differences among them, the core idea of relational accounts is

when this has been specified. The main terms are reported in the glossary (Appendix 3) and schematised in Fig. 2.

⁶ Admittedly, this formulation entails some circularity given that the notion of moral autonomy is underdefined in the literature.

⁷ The expression "contemporary accounts" refers to the philosophical theories and models of autonomy developed starting from the twentieth century, especially in the liberal tradition. These accounts shift the focus from moral to personal autonomy. Different contemporary accounts are positioned along the two axes highlighted in the text (internalism/externalism, proceduralism/substantivism) in more or less common combinations (e.g., internalism and proceduralism).

that personal autonomy is caused or constituted by relational conditions such as social support, social recognition, reciprocity, etc. For instance, Mhlambi and Tiribelli [7] argue for reformulating the AI ethics' principle of autonomy by drawing on both Western feminism and non-Western Ubuntu philosophy. Following the latter, personal autonomy is grounded in interconnectedness, requires relatedness, reciprocity, and active participation, and implies duties toward others [7].

This relational trend partly overlaps with the tendencies to provide integrated and/or multidimensional accounts of personal autonomy. The following four publications can be read in this light since they draw on various contemporary accounts to develop rather different conceptions of autonomy and analyses of AI's impact. For instance, Bonicalzi et al. [25] draw on the internalist/externalist distinction to highlight different effects of recommender systems on *Descriptive Autonomy* (see Sect. 4): manipulation can affect both internalist and externalist conditions of autonomy, reshaping identity affects internalist authenticity, and filtering information affects externalist conditions. An alternative strategy consists in overcoming and reconciling the traditional distinctions between contemporary accounts. For instance, Formosa [29] interprets autonomy as comprising proceduralist, substantivist, and externalist elements: a range of competencies, authentic values, and sufficient available options. Similarly, Rubel et al. [39] propose an *ecumenical view* (an account compatible with a broader range of views) by integrating what they call "psychological autonomy" (corresponding to internalist-proceduralist accounts) and "personal autonomy" (corresponding to socio-relational accounts). Their account thus includes both procedural independence (competency and authenticity conditions) and substantive independence (social conditions such as valuable options and self-respect). Likewise, the *pragmatic account* by del Valle and Lara [53] incorporates relational elements into a revised procedural autonomy where competency (i.e., information management and deliberative skills) and authenticity conditions map onto two different sets of problems raised by AI: on the one hand, manipulation and cognitive dependence, on the other, challenges to reflective endorsement, authenticity, and explainability.

Similarly, an increasing number of papers (e.g., [3, 23, 31, 42, 54–56]), develop relatively new accounts of human autonomy, sometimes called *multidimensional*, comprising multiple components and dimensions.⁸ For this reason, these

seven papers [3, 23, 31, 42, 54–56] are grouped together, although not all of them use the label "multidimensional". However, this group is internally heterogeneous due to the disagreement on the number, type, and definition of components or dimensions. Moreover, while some of these components and dimensions are derived from contemporary accounts, they can be called or defined in different ways. The followings are some illustrative examples (see Table 2 for a more detailed description). Drawing on the contemporary philosophical literature on personal autonomy, Prunkl [3] proposes a framework that includes and distinguishes between two main dimensions of autonomy: *authenticity* or the internal dimension (i.e., capacity to decide and act on the basis of one's own motives) and *agency* or the external dimension (i.e., effective capacity of making choices and enacting them). According to Prunkl [3], this distinction allows us to highlight two different sets of concerns about AI's impact on human autonomy: while manipulation affects authenticity, the challenges to human control affect agency (see Sect. 4). Multiple dimensions or components characterise the "maximally informative account" by Laitinen and Sahlgren [23], including capacities for self-determination, normative requirements, respect and self-respect, performativity or exercise, and external resources (called "sociotechnical bases"). The aim is to provide a useful framework even if one doesn't recognise all these aspects of autonomy to assess the impact of AI on both individual components and external bases. A bold attempt to rethink human autonomy is the multidimensional and relational model by Savolainen and Ruckenstein [54], showing that autonomous agency is mediated and co-shaped by algorithms. For this purpose, they distinguish four dimensions of human autonomy in algorithmic relations: two pertain to an *instrumental horizon* (i.e., algorithms are technical procedures for ordering social life) and consist of *algorithmic capacity* (i.e., capacity to identify and understand algorithms) and *situational mastery* (i.e., using algorithmic affordances for one's goals). The other two pertain to an *intimate horizon* (i.e., algorithms give rise to affective relationships that involve sharing a personal space) and consist of *breathing space* (i.e., room for realisation and reflexivity) and *co-evolution* (i.e., self-understanding through algorithms).

3.2.3 Psychological conceptions of personal autonomy

The analysis by Calvo et al. [58] stands out from the rest of the literature for referring to the psychological account of autonomy derived from the Self-Determination Theory [59]. In this theory, autonomy is a basic psychological

⁸ The term "dimensions" comes from Mackenzie's relational account of personal autonomy [57] and is relatively common in the reviewed literature. However, despite being an established term, there isn't a standard definition specifying and listing these dimensions. As shown in this subsection, this vagueness is reflected in the reviewed literature and explains the heterogeneity of multidimensional accounts. Due to the lack of a shared terminology in the literature, components and

dimensions can include several things such as constitutive aspects of autonomy, forms that autonomy can take, as well as elements that are not part of autonomy but preconditions for it.

need for well-being that expresses the self-endorsement of motives and behaviours. According to Calvo et al. [58], human autonomy requires: a feeling of willingness, volition, endorsement; lack of pressure and control; lack of deception and misinformation. As specified by the authors, this is a “working definition” of autonomy since it’s not an established and complete conceptualisation but a sufficient basis for addressing several issues raised in the context of AI. This framework is used to highlight both problems (e.g., coercion, manipulation, surveillance, addiction) and benefits (e.g., help with information overload) of AI for human autonomy. More precisely, Calvo et al. [58] link the Self-Determination Theory with the “Motivation, Engagement and Thriving in User Experience” model to assess different experiences and issues of autonomy in six “spheres of technology experience” (adoption, interface, task, behaviour, life, society). For instance, this reveals that a user can feel autonomous at the level of the interface (e.g., self-tracking) but not at the level of task (e.g., step counting). In the same vein, Shevlin [60] suggests combining Self-Determination Theory and the Motivation, Engagement and Thriving in User Experience model for the assessment and ethical development of social AI. The Self-Determination Theory is also the framework used by Varshney [33] to argue for the operationalisation of the principle of respect for human autonomy in recommender systems.

3.2.4 Bioethical conceptions of patient autonomy and other context-related conceptions

Given that autonomy is context-dependent, its conceptualisation, conditions, and implications also vary according to the context (see also [61]). Consequently, the bioethical principle of patient autonomy is paramount in the debate on AI applications in healthcare (e.g. [62–66]). In bioethics, patient autonomy is the capacity and right to make medical decisions aligned with one’s values and goals. This traditionally requires intentionality of choice, understanding of relevant information, and independence from controlling influences [67]. As it will be shown (Sect. 4.2.8), these conditions can be either supported or undermined by AI. For these reasons, Laacke et al. [65] argue that the traditional principle of patient autonomy must be complemented by the elements of optionality and engagement. Fewer analyses ([32, 68, 69]) provide examples of the impact of workplace automation on individual autonomy. Among them, Aizenberg et al. [69] criticise algorithmic hiring assessment by developing the rather new conception of autonomy over self-representation: the ability to present one’s identity, experiences and aspirations, requiring control over the shared information and engagement with the world (see Sect. 4).

3.3 Discussion

The literature doesn’t provide a shared definition, but various alternative conceptions of human autonomy. This is not surprising but reflects the complexity and contextuality that has always characterised this concept in the previous philosophical literature (see [15]). Although some heterogeneity in the conception of autonomy is inevitable and enriching, the lack of a common framework undermines a comprehensive understanding of the debate.

Amidst this heterogeneity, the literature also presents some more recurrent notions and trends that are here summarised. The prevailing category is personal or individual autonomy, consisting in the capacity to decide and act in accordance with one’s motives and goals (e.g., [3, 23, 29]). The focus on personal autonomy is both in line with previous autonomy accounts and debates in contemporary social and political philosophy (i.e., from the twentieth century), as well as motivated by the current impact of AI on various aspects of individual life involving personal choices. Unsurprisingly, moral autonomy is mostly addressed in debates around the use of AI for ethical decision-making. The relevance of personal and moral autonomy in the literature is also partly explained by the focus of the review on AI ethics’ academic debate, which excludes contributions from other literatures where different categories of autonomy are employed. Therefore, one could wonder whether other, either pre-existing or new categories of autonomy are required in this debate. Clearly, an advantage of developing new categories of autonomy is that they would be more tailored to the context of human-AI interaction than traditional ones. However, the worry is that additional categories might further inflate and complicate the concept of autonomy and, consequently, the assessment of AI’s impact on it.

One could also ask whether the traditional distinction between moral and personal autonomy still holds in AI ethics, or whether the two are instances of the same category. Indeed, the interpretation of moral autonomy emerging from the literature departs from Kant’s definition and comes closer to personal autonomy: the capacity to give a universal rule to oneself is replaced with the capacity to form and act on one’s moral values. This also raises the question whether the traditional Kantian notion of moral autonomy is still relevant to the current debate on autonomy in AI ethics. On the one hand, Kantian accounts in the reviewed literature are few and quite different from one another, suggesting that the traditional version of Kant’s theory of autonomy plays a minor role in this debate. On the other hand, these accounts show that some core aspects of Kant’s theory of autonomy are relevant to discuss the impact of AI. Taken together, these opposite results suggest that a Kantian account of autonomy remains useful in addressing different issues

raised by AI (see Sects. 3.2.1 and 3.2.2; see also Sects. 4.2.6 and 4.2.7 for further discussion), once revised or adapted to debates in AI ethics.

More often, the literature is influenced by several contemporary accounts since they turn useful for analysing the impact of AI on various conditions of personal autonomy. While there are papers that adopt the theories that better serve their analysis (e.g., Raz's self-authorship), there is also the tendency to integrate different accounts, especially relational ones. The latter strategy allows for a more comprehensive assessment of AI's impact on personal autonomy than what can be achieved by a single account alone.

In line with a traditional understanding of individual autonomy, the main aspect is rational deliberation and decision-making. However, there is also an increasing recognition of the multidimensional and relational nature of personal autonomy. This tendency suggests that traditional individualistic accounts are perceived as inadequate to address the multifaceted impact of AI on human autonomy. By contrast, its assessment should also consider external (especially, socio-relational) factors. For these reasons, multidimensional and relational accounts represent an underexplored but promising starting point for a more comprehensive and nuanced understanding of human autonomy and its relation with AI.

The emergence of relational accounts of autonomy in the reviewed literature is in line with a broader turn in AI ethics: from traditional approaches (e.g., principlism) and individualist concepts (e.g., personhood, privacy) toward relational, as well as decolonial approaches (see [70] for an overview of relational AI ethics; see [71–73] for approaches to technology based on Confucianism and Buddhism; see also the *relational understanding* by [74]). Therefore, relational accounts of autonomy arise in response to the limitations of traditional Western individualistic accounts (e.g., Kantian). The latter are not comprehensive or inclusive enough to address how AI impacts the relational conditions of autonomy (from the integration of AI in decision-making to the power relations within a sociotechnical system; see e.g. [7, 54]), especially of those mostly affected by the deployment of such systems, such as marginalised communities (see [7]). For instance, Tiribelli [50] criticises the literature on health recommender systems for identifying autonomy with “informational self-determination” (i.e., the capacity to make better and independent choices and to be in control through information), which overlooks the socio-relational conditions of autonomy. A similar worry can be raised in the context of AI-based neurotechnologies, where individual autonomy is intertwined with “mental self-determination” and “cognitive liberty” (see [11] for an explanation of these concepts). As recalled, Mhlambi and Tiribelli [7] argue for a relational turn in reformulating the AI ethics' principle

of autonomy by appealing to both feminist and Ubuntu philosophy.

Therefore, to further develop relational accounts, the literature would benefit from greater engagement with non-Western philosophical traditions, like Ubuntu, Confucianism, or Buddhism. As shown in the previous Section, the reviewed literature almost exclusively refers to conceptions of autonomy developed in the Western philosophical tradition. This is partly explained by the centrality of autonomy in Western philosophy. However, although various non-Western traditions might not discuss autonomy in the same terms of Western philosophical accounts—especially individualistic ones—they can offer alternative conceptualisations and discussions on related topics (see e.g., [71, 73, 75] for examples and further discussion). For instance, some non-Western philosophies contribute to relational autonomy also through their relational conception of personhood, identity, and choice (see e.g., [73]). This confirms that AI ethics literature needs to engage more with non-traditional conceptions of autonomy for a more comprehensive understanding of autonomy and its relation with AI systems.⁹

This revisionist trend partly overlaps with accounts that merge different approaches (e.g., procedural and relational) and/or that develop multidimensional accounts. An illustrative example is the pragmatic account by del Valle and Lara [53]. As stated by the authors, their proposal follows the pragmatic turn ([76, 77]) informed by the instrumental value of concepts, the empirical confirmation, and the rejection of traditional dichotomies [53]. As mentioned, multidimensional accounts arise in response to a revisionist need as well: a more comprehensive and fine-grained understanding of human autonomy and assessment of AI's impact on its multiple dimensions. Consequently, multidimensional accounts provide theoretical frameworks under which different impacts of AI on human autonomy can be spelt out and addressed. However, the variety in terms of which and how many dimensions should be included can further complicate an already complex concept and landscape. This suggests that more engagement and coordination within different contributions in the literature could help avoid this drawback.

Together with revising and expanding philosophical accounts, the literature would also benefit from integrating the insights of psychological theories of autonomy. As shown by the examples in the literature (Sect. 3.2.3), the benefit of psychological accounts is to provide the means for operationalising the concept of autonomy and measuring how it is affected by AI in real-case scenarios, as well as for more efficiently designing AI systems for human autonomy.

⁹ I thank an anonymous reviewer for the insights and references on relational and non-Western approaches in AI ethics discussed in this and previous paragraphs.

Lastly, the context of analysis also matters to define the conditions and implications of autonomy. For this reason, a single account of autonomy applicable to different contexts is difficult if not counterproductive. This goes in favour of maintaining a conception of autonomy that is heterogeneous, flexible, and adaptable to different scenarios. However, one might worry that developing different specific accounts of autonomy for each context would excessively inflate this concept and undermine a comprehensive and systematic assessment of AI's impact on human autonomy. All this reveals that the main conceptual challenge consists in rethinking and operationalising the concept of human autonomy in its relation with AI.

4 The impact of AI on human autonomy

4.1 Background

Traditional challenges to autonomy range from meta-physical issues (e.g., determinism), to psychological limits (e.g., weaknesses of the will), to external influence (e.g. manipulation, coercion, paternalism) and oppression (e.g., surveillance, marginalisation) (see [15] for an overview). Technological development has always brought about both challenges and opportunities for human autonomy (as in the cases of reproductive technologies and surveillance technologies). Against this background, this Section analyses the multi-faceted impact of AI on human autonomy. Since the reviewed literature is almost exclusively concerned with the impact on the exercise of autonomy, this will be the focus of this Section. However, some space will also be given to the implications for the conception of autonomy.

4.2 Findings

The assessment of AI's impact on human autonomy is not homogeneous but depends on the technology, context, and conception of autonomy. Therefore, the literature is divided between stressing the challenges and recognising the opportunities, although it is more focused on the formers. Moreover, the debate often concerns the impact on individual or personal autonomy, with a focus on the effects on autonomous decision-making (see Sect. 3). To organise its results, the following overview distinguishes between two main sources and modalities of AI's impact on human autonomy.¹⁰

¹⁰ This is not the first and only possible overview and categorisation of AI's impact on human autonomy. The distinction proposed here partly builds on the frameworks developed by Prunkl [2, 3, 78] and integrates them by identifying and classifying the sub-categories of issues included under two sources of AI's impact. For alternative overviews and categorisations, see e.g., Laitinen and Sahlgren [23], and Savolainen and Ruckenstein [54].

The first consists in AI mediating human deliberation, decision-making, and behaviour. This means that interacting with AI influences the individual capacity to form one's motives, to make one's decisions, and to behave accordingly. A major theme concerning about half of the literature is the influence of AI on the individual capacity to deliberate on one's motives and to decide and act accordingly. The literature is divided between recognising both negative and positive influences. On the one hand, AI can negatively interfere with decision-making, critical thinking, preference formation, personal identity, and behaviour (e.g., [25, 41, 42, 79–82]). On the other, AI can support or enhance autonomous deliberation and decision-making by improving decisional competencies and favouring authentic choices (e.g., [35, 42, 83, 84]). Both cases affect several conditions of autonomy: competency (bypassing or improving reasoning and deliberative skills), authenticity (undermining or favouring deep preferences), understanding (targeted or opaque information), and independence (undue or supportive external influence). The issues of AI mediation are raised by various kinds of AI-based decisional assistants, especially recommender systems.

The second consists in AI automating and replacing human decisions and actions. This means that outsourcing tasks to AI affects the conditions for making choices and for acting in line with one's values and goals. Differently from previous cases of technological automation, AI affects human autonomy by replacing both physical and cognitive tasks. For this reason, Danaher [26, 27, 52] considers AI assistance as a special form of automation. It consists in "algorithmic cognitive outsourcing", defined as the process of "offloading a cognitive task to a smart algorithm" [26]. This is a quite broad category that can include several levels of AI-based automation, especially because the boundaries between AI assistance and replacement are not always clear-cut. For this reason, some issues raised by algorithmic outsourcing (e.g., manipulation from AI assistants) overlap with AI mediation. Overall, there is a trade-off between negative and positive effects of AI automation, as well as between different components and conditions of autonomy (see [56]): AI automation can restrict human control over decisions (i.e., autonomy as the capacity to make choices) but increase the range of viable options (i.e., autonomy as the capacity to enact one's choices and goals). In both cases, automation mostly affects the conditions of agency (reducing or expanding), optionality (decreasing or increasing options), competency (deskilling or upskilling) and authenticity (implementing or substituting choices, values, and goals). Beyond these general remarks, the impact of AI automation varies depending on the technology discussed in each case, ranging from autonomous vehicles to decisional assistants.

Following similar frameworks (cf. [2, 3, 78]), the distinction between AI mediation and automation shows how different kinds of AI's impact map onto different aspects of autonomy: while AI mediation or influence mostly affects the internal deliberative and decisional component, AI automation and replacement mostly affect the external and practical component of autonomy. However, the classification here proposed is not meant to be entirely rigid. Clearly, human-AI interaction always involves some level or form of both mediation and automation. Some issues can result from both AI mediation and AI automation, and can target multiple aspects of autonomy at the same time. Moreover, the impact of AI mediation and AI automation on autonomy—and the related debates—also depends on the technology and the context at stake. While the impact on human autonomy is discussed in relation to a wide range of AI-based technologies, recommender systems are among the most illustrative examples of the problematic influence on autonomous decision-making. Therefore, the following analysis will present the main clusters of issues by linking them with the technological developments and/or contexts that are most illustrative of these debates. Lastly, AI's impact also depends on the role played by the broader network of relations that shape the development and deployment of AI, rather than on the interaction with AI alone.¹¹ With these caveats, several different themes discussed in the literature can be traced back to the two broad categories here proposed. Therefore, the following overview represents a compass for mapping and classifying different ways in which AI affects human autonomy. The following subsections provide a more detailed analysis of the issues discussed in the literature, while the results of this categorisation are also summarised in Table 3 of Appendix 2.

4.2.1 AI mediation: manipulation, deception, coercion

Some forms of influence, such as coercion and manipulation, are traditionally considered a major threat to autonomy [15]. AI reshapes and amplifies these threats in terms of methods (targeted techniques), scale (more widespread, more pervasive, more personalised), scope (public and private sphere), and power relations (both public institutions and private companies). The novelty of AI's influence is explained by Floridi [87] with the concept of “hypersuasion”, indicating the unprecedented power of AI to exert influence through personalized and data-driven methods. According to Floridi [87], hypersuasion is based on two factors: first, the capacity of ML to process huge amounts

of data to extract information that can be used for persuasive purposes; second, and more recently, by the capacity of Generative AI to automate the production of human-like and targeted content. Therefore, hypersuasion can be seen as a broad category comprising several forms of AI's influence discussed in the following subsections.

Among them, the main issue in the reviewed literature is *manipulation* (e.g., [3, 11, 23, 25, 26, 42, 60, 88]). Manipulation broadly consists in a form of intentional, indirect, and covert influence that bypasses reasoning and exploits decisional weaknesses to favour an external party (see [11] for a recent overview). However, the definition, features, and scope of manipulation (especially online) are object to disagreement (see [89] for a discussion on the philosophy of online manipulation). Nevertheless, online and AI manipulation often occurs by shaping the choice environment in a way that prevents rational deliberation and authentic choices [90]. Several AI applications (e.g., social media algorithms, chatbots, deepfakes) raise manipulation issues [11, 91], especially in the case of recommender systems: by exploiting decisional heuristics and biases, recommender systems vehiculate users' choices toward items they wouldn't have chosen otherwise [25]. The risk of manipulation is increased by the relationships with conversational and relational AI that makes recommendations and advices to influence users for commercial purposes [60, 85]. The recent development and deployment of Generative AI can increase the quantity and quality of manipulation (e.g., more targeted content; [87]), including emotional manipulation (e.g., from Generative AI-based chatbots [85]).¹² Overall, AI manipulation threatens autonomy both because of its content (misaligned with user's preferences) and its method (disrespectful of deliberative capacities) [25]. Moreover, manipulation undermines mental self-determination because consciousness and decisions are controlled or influenced by others [11].

Despite widespread concerns, there is also disagreement on whether AI-based manipulation is always detrimental to autonomy. Indeed, manipulation can have different effects on different aspects of autonomy. As suggested by Bonicalzi et al. [25], manipulation can compromise Descriptive Autonomy without causing a loss of Normative Autonomy (e.g., undermining the capacity to make informed decisions without infringing the right to be treated as an end), or can violate Normative Autonomy without compromising Descriptive Autonomy (e.g., being targeted by a manipulative attempt but being able to spot and avoid it). Moreover, the tri-partite model of autonomy proposed by Sass

¹¹ In this respect, several violations of autonomy are related to the lack of user's informed consent to the development or use of AI. Emblematic cases are manipulative commercial recommendations [60, 85] and the non-consensual development of digital duplicates [86].

¹² A reviewer pointed out the possible introduction of advertising created by ChatGPT. Given that this is such a recent development, this possibility has not yet been discussed in the academic literature, but would be an interesting and relevant development for the debate on AI manipulation.

[42] distinguishes the effects of manipulation on different components of autonomy: for instance, manipulation can improve overall autonomy, despite reducing *control* over decision-making (i.e., awareness of available choices and approval of external influences), by improving *agency* (i.e., capacity to take action and develop skills) and/or *authenticity* (i.e., the result of a process in which the individual has several preferences, a developed life plan, a set of capacities for critical reflection that can exercise them to select preferences); vice versa, manipulation can diminish overall autonomy, despite increasing control, by reducing agency and/or authenticity.

Partly overlapping with manipulation, AI *deception* interferes with decision-making by instilling false beliefs [23] through e.g., personalised, targeted, and attention-catching information [53, 92]. A different, concerning, and controversial case of manipulative and deceptive influence comes from chatbots and social robots that trick users about their human-like features, mental states, or capacities, especially if powered by Generative AI [93]. Although not all cases of anthropomorphism need to be problematic, this kind of deception favours false beliefs or emotional dependence that enable manipulation of users' choices against their influence.

Although less discussed in the literature, *coercion* is a traditional kind of undue influence possible also with AI. Coercion undermines autonomy not by manipulating decision-making but by removing freedom of choice due to a restricted range of valuable options [90]. More precisely, online coercion often acts by making something an unavoidable precondition for something else (e.g., accepting terms and conditions or watching mandatory ads) [23, 92]. This suggests that identifying and assessing AI-based forms of coercion requires taking broader power relations between companies and individual users into account.

4.2.2 AI mediation: nudging and assistance

Recommender systems and other AI-based decisional assistants also raise questions about the more subtle and controversial case of *nudging*: employing techniques that shape the choice architecture to influence behaviour toward better choices [94].¹³ Whether nudges (analogical or digital) threaten or support autonomy is still an open question [84,

90, 95]: while nudging techniques are advocated for preserving freedom of choice [94], they are criticised for being manipulative, bypassing rational decision-making (see [13] for an overview), and not developing autonomy competencies [96]. Digital and AI nudges raise further problems: while sharing core elements of traditional nudges (e.g., altering the choice architecture by exploiting cognitive biases), they are personalised, pervasive, irresistible, and hidden in opaque algorithms [84]. Moreover, Yeung [97] theorises a new and problematic form of digital nudging, called “hypernudge”: digital techniques aimed at channeling decision-making, characterised by being networked, constantly updated, dynamic, pervasive, and personalised. According to the author, these characteristics make hypernudges a challenge for freedom of choice. Additionally, Lanzing [98] argues that hypernudging threatens autonomy also through privacy violations: by hiding reasons and agents behind choices and by reducing options, hypernudging compromises user's control over whom and how their data will be used (informational privacy), as well as over users' decisional process (decisional privacy).¹⁴ For all these reasons, AI nudges and especially hypernudges, could be seen as either a new version of nudges or a new and different phenomenon.

The debate on nudges has also been fuelled by proposals to develop Artificial Moral Advisors (henceforth, “AMAs”): AI systems designed to help with ethical decision-making by providing advice or recommendations that enhance the capacity to decide in line with one's moral values, principles, and goals [34–36] (see also [46] for a literature review). According to Giubilini and Savulescu [34], such AI assistants can improve ethical decision-making by either offering support in identifying the best (i.e., most morally aligned) choice or by improving moral deliberation skills. In their proposal, the AMA provides users with recommendations that allow them to always act in accordance with their moral goals. However, this proposal is criticised by Lara [35] because users are treated as recipients of AMA's indications, not truly engaged with the deliberative process but relegated to a marginal role (e.g., choosing the initial values, approving the output). For this reason, the decisional process is driven by the algorithm alone and not informed by the user's values. Similar systems thus threaten human autonomy in three ways: by replacing humans in ethical decision-making, by pushing them toward the best choice,

¹³ The concept of “nudge” was developed by Thaler and Sunstein, and defined as: “any aspect of the choice architecture that alters people's behaviour in a predictable way without forbidding any options or significantly changing their economic incentives” [94]. Since humans are not perfect rational decision-makers, nudges help them make better choices (i.e., as if they were fully rational and informed). For this purpose, nudging techniques shape the choice architecture in ways that trigger automatic cognitive processes (e.g., heuristics and biases) without removing options or changing costs to preserve freedom of choice.

¹⁴ According to Lanzing [98], both informational and decisional privacy are relevant for autonomy. More precisely, decisional privacy supports decisional autonomy by granting control and protection over the ways we make judgments and decisions against unwanted access and interference (see [99, 100] on the right to decisional privacy). By contrast, a lack of decisional privacy makes one vulnerable to unwanted external interferences with one's decision-making process that influence one's behaviour [23].

or by advising them according to the AMA's beliefs [35]. To counter these issues, Lara and Deckers [36] propose "SocrAI", a conversational bot designed for the moral education of the user. Rather than giving moral advice, it teaches users to better reason about moral issues in an independent way [35]. For this purpose, Lara and Rodríguez-López [84] propose to implement nudges that are informative, transparent, and resistible. More precisely, they define "Socratic nudges" as information or indications provided by the artificial assistant to counterbalance some biases that impair rational deliberation. This kind of moral education resembles the concept of "boosts" proposed in behavioural economics: techniques that aim at changing behaviour for better decisions by improving or creating competencies [101]. Although this concept is not explored in the reviewed literature, it could help design AI assistants that support human decisional autonomy.

The debate on AMAs suggests that AI's influence through decisional assistance could help overcome some of the internal limits of autonomy (e.g., weaknesses of the will and cognitive biases). In this line, Kempt et al. [83] argue that shifting the focus from individual to relational autonomy unveils other ways in which the interaction with and influence of AI supports decision-making. More precisely, they identify three ways in which human-AI relationships can support autonomy: first, improving decision-making skills by cognitively enhancing our understanding of different available options; second, allowing for reflective moments to appreciate the difference between one's deep preferences and those coming from the social context; third, increasing the relations that influence and improve decision-making.

4.2.3 AI mediation: algorithmic framing

Algorithmic recommendations are also illustrative of other forms of influence on decisional autonomy, not completely reducible to the previous categories and specifically related to their filtering, prediction, and framing of information. For all these reasons, the influence of algorithmic recommendations deserves a separate categorisation and discussion.

In principle, personalised recommendations should support decisional autonomy by leading to decisions aligned with one's preferences thanks to more relevant options and information, against so-called "informational overload" (for an explanation of the phenomenon see [102]). However, algorithmic recommendations have several problematic implications for individual autonomy. For instance, Bonicalzi et al. [25] identify three ways in which recommender systems affect Descriptive Autonomy, which will be covered in this subsection: manipulating decisions, shaping personal identity, affecting critical thinking.

A first issue is that algorithmic recommendations do not properly reflect users' preferences. Rather, they steer users toward choices they wouldn't endorse otherwise, usually for external interests [25, 37, 79]. As already mentioned, this can constitute a form of manipulation that undermines Descriptive Autonomy, but also a violation of Normative Autonomy by treating users as mere means [25]. This form of influence is related to so-called "transformative effects" [41] of algorithmic recommendations: our conceptualisation, ontologisation, and understanding of the world are changed and shaped by reducing the number and the diversity of available options, while showing only what is relevant for the algorithm's criteria. According to Mittlestadt et al. [41], this is the "paradox" of recommender systems: while they should improve users' decision-making by providing only relevant information, they actually undermine it by pushing toward options that hardly reflect authentic preferences. This paradoxical situation is explained by the tri-partite model proposed by Sass [42] showing that the influence of recommender systems has different effects on different dimensions of autonomy (Sect. 4.2.1).

However, the personalised filtering of options is linked to a second related problem: algorithmic recommendations can shape one's preferences by reinforcing and perpetuating past choices [25, 37, 79]. This can lead to phenomena like "filter bubbles" [103] that prevent exposure to different information. For similar reasons, algorithmic recommendations can cause "algorithmic predeterminism" [38]: the user's choice is either already known or steered by the algorithm toward the predicted option. According to Tiribelli [38], this interferes with moral autonomy (i.e., the power to form a moral disposition and identity), by surpassing the reflective endorsement of motives of choice.

For similar reasons, recommender systems can also threaten personal autonomy by shaping personal identity [25, 79] since the latter provides an internal coherence that overlaps with authenticity conditions of personal autonomy [25]. The reason is that recommender systems classify users in social categories that don't reflect their self-perception [25]. Similarly, Laitinen and Sahlgren [23] discuss the possibility that AI violates human autonomy through misrecognition and misrepresentation of one's identity (e.g., wrong classification).

Lastly, recommender systems can exclude potentially relevant information and hide the reasons behind recommendations [33, 88]. The opacity of algorithmic decision-making prevents the understanding of explanations that is necessary to reject or integrate such recommendations into one's autonomous decision-making [88], as well as to shape future outcomes to realise one's goals [32].

All these phenomena interfere not only with decision-making but also with critical thinking: indeed, AI can

mediate and influence knowledge acquisition and belief formation through filtering, standardisation, misinformation, etc. [25]. In conclusion, filtering and personalisation can reproduce and amplify but also overcome some of the traditional threats to autonomy: they can create new forms of determinism and heteronomy, stress and exploit internal limitations (e.g., lack of information, cognitive and decisional biases, inconsistent preferences, etc.), but also overcome some of them (e.g., relevant information, preference prediction, etc.).

4.2.4 AI mediation: surveillance

With the advancement of big data combined with AI computational capacities, other forms of indirect influence consist in data sharing, analysis, and decoding. This can affect rational and competent deliberation by causing heteronomous influence, approximation and standardisation, and incomprehensibility of choices [104].

Algorithmic data analysis also increases risks of *surveillance*, that threatens the capacity to decide and act autonomously due to the feeling of being constantly observed and controlled [105]. More precisely, Danaher [52] warns against a subtle and pervasive kind of surveillance, which he calls “micro-domination” (the concept is derived from [106]). Differently from traditional forms of domination (e.g., slavery), algorithmic domination consists in the accumulation of small cases of daily monitoring and tracking that make algorithmic control unescapable: hiding certain information becomes impossible due to the continuous sharing of personal data among different but interconnected devices, platforms, and apps [52]. The unprecedented level of monitoring and surveillance enabled by AI doesn’t only concern personal/private life but also the workplace, where it can lead to a loss of workers’ autonomy [68]. Algorithmic surveillance is also linked to the widespread adoption of various technologies for tracking and monitoring (especially after the COVID-19 pandemic) [107]. Similar cases of algorithmic monitoring and surveillance are analysed by Ho [107] through a relational perspective: addressing the socio-relational contexts that underpin AI healthcare monitoring reveals hidden forms of surveillance and privacy violations that undermine individual self-determination.

4.2.5 AI automation: enabling or disabling decisions and goals

The debate on AVs (i.e., Autonomous Vehicles) illustrates how AI automation can have opposite effects on different dimensions of autonomy. As argued by Fossa [8], this debate has focused on automating moral decisions through crash-optimisation algorithms: such technologies can enhance or

reduce moral autonomy by enlarging or restricting the range of available ethical decisions. However, Fossa [8] also shows that the impact of CAVs (i.e., Connected and Automated Vehicles) on human autonomy is broader: on the one hand, the main threat consists in bypassing judgment, decision, and action (e.g., automated route recommendation); on the other, partial automation can improve driving decisions, while full automation can provide more opportunities (e.g., free time, route options, mobility choices, etc.). Therefore, the impact of AVs depends on the level of automation, as well as on an internal tension between two conceptions of autonomy [56, 108]. In the multidimensional account by Santoni de Sio and Fossa [56], full automation can reduce autonomy as “self-determination of driving decisions”, but improve “freedom to pursue a good life through mobility”.

Further ways in which human-AI interaction can enhance or diminish human autonomy are spelt out by Formosa [29] in the case of social robots. First, outsourcing tasks to social robots can either favour or prevent the achievement of valuable ends ourselves. Second, social robots can help or undermine authentic choices. Third, social robots can improve or decrease our autonomy competencies. Therefore, while offloading irrelevant tasks to social robots liberates resources for more important ends, offloading valuable tasks prevents us from making decisions based on our values and goals and leads to a deterioration of decisional competencies [29]. A similar analysis can be applied to the more recent cases of Advanced AI Assistants [93] and so-called “digital duplicates” [86], when used to decide and act on one’s behalf. However, partly due to the novelty of such technologies, their impacts on human autonomy have not been extensively analysed yet.¹⁵

4.2.6 AI automation: deskilling and dependence

Technological automation is associated with the potential degeneration of capacities delegated to technologies and no longer exercised [109]. Similarly, *deskilling* is among the issues raised by digital or AI assistants, although the implications for autonomy are not always immediate but need to be spelt out.

According to Danaher [26], the concern for autonomy raised by AI outsourcing is a version of the so-called “degeneration argument” by Carr [109]. Against it, Danaher [26] argues that the damaging effects of deskilling depend on the commonality of and cognitive resiliency required by the task, as well as by the benefits of delegation and the value of the task. This analysis is followed by Laitinen and Sahlgren [23] in relation to what they call “cognitive

¹⁵ In these cases, concerns for autonomy are not limited to the interaction but arise at different stages, starting with the risk of bypassing consent for the creation and deployment of the digital duplicate [86].

heteronomy”, resulting from the degeneration of cognitive capacities necessary for autonomy (see Sect. 3.2.2). In a similar vein, Aylsworth and Castro [48] point out that losing capacities as a result of outsourcing cognitive tasks to AI is problematic only if it affects capacities that are relevant for the development or exercise of autonomy. In a more negative vein, Lu [80] argues that algorithmic dependence and deskilling is an inevitable challenge of automated decision systems, because offloading tasks doesn’t liberate mental resources but deteriorates skills. By contrast, Heinrichs [55] argues that overemphasising the risks of digital assistants for a “passive” component of autonomy (consisting in self-perfection and corresponding to authenticity) overlooks the potential gains for an “active” component (consisting in self-knowledge and corresponding to competency).

Regarding moral deskilling, the major contribution is offered by Vallor [110] who warns against the possible atrophy or loss of moral skills and virtues resulting from offloading moral decisions to AI (especially, Autonomous Weapon Systems). Although not explicitly stated, the loss of moral skills would undermine moral autonomy by weakening the capacity to make moral decisions.

The risk of deskilling is also linked to another problem of AI outsourcing, namely *cognitive and emotional dependence*. While inter-dependence is compatible with or constitutive of relational autonomy, some dependency relations are autonomy-undermining. For instance, Formosa [29] argues that depending on a social robot to make decisions and actions can decrease trust in one’s capacities and dissipate one’s skills. Relatedly, he suggests that over-reliance on social robots makes autonomy more vulnerable since e.g., we are not able to achieve our ends in case of technological malfunctioning. Moreover, human-AI interaction can also create emotional dependence on technologies. Recently, the development of Generative AI-based chatbots is increasing the concerns for both cognitive and emotional dependence. A clear example is provided by Lindemann [85] in the case of griefbots (i.e., AI-based digital recreation of a dead person): scaffolding grief to griefbots produces emotional dependence since the user cannot stop using the bot for the fear of losing the relationship with the deceased. Additionally, Lindeman [85] stresses that this emotional dependence can be exploited by external parties to promote commercial products against users’ interests and goals. Therefore, emotional dependence doesn’t only undermine autonomy in itself but also makes it more vulnerable to other threats, especially manipulation. This is additional proof that, if independence means freedom from manipulation, algorithmic outsourcing raises unique threats to autonomy [26].

4.2.7 AI automation: heteronomy

Various cases of automated decision-making (from AVs to algorithms for risk allocation and hiring assessment) raise issues connected to loss of autonomy. Although rarely expressed in these terms, the literature is also concerned with the risk of turning AI automation into a form of *heteronomy* (i.e., being subjected to something else rather than being governed by one’s own rules, norms, values, etc.). According to Laitinen and Sahlgren [23], cognitive heteronomy raises two questions: whether we defer to AI in non-autonomous ways and whether such deference reduces our capacity to be the authors of our lives. To answer these questions, they follow Danaher’s analysis [26]: first, outsourcing is an autonomous action if the decision to delegate is not coerced/manipulated, is limited to certain tasks, and provides some benefits (e.g., more resources for autonomous choices/actions); second, the risks of degeneration depend on its diffusiveness, on the value of the task, and on the benefits of delegation.

AI-based heteronomy affects both the autonomy of the agent who should make the decision and the autonomy of the agent who receives an automated decision. The first case is explained by Chiodo [24, 45] in terms of “overturning” human autonomy (Kantian self-rule): while technologies acquire more machine autonomy, humans become more automated, meaning that they are not guided by a self-given law but by something external and arbitrary. The second case is discussed by Rubel et al. [39] in relation to automated decision systems that don’t pass the “Reasonable Endorsement Test” for algorithmic assessment. This means that automated decisions violate autonomy by subjecting people to reasons and principles that not everyone would reasonably endorse [39].¹⁶

4.2.8 The contextual impact of AI

Although the issues analysed in the previous sub-sections occur in several scenarios, AI’s impact on human autonomy can vary depending on the context and the account of autonomy adopted in each case. For this reason, it is important to look at what issues are raised in each context where AI is employed. For instance, issues of AI mediation arise in online environments such as social media, commercial platforms, and apps (e.g., [3, 25, 58, 79]).

A case illustrating various issues previously discussed is the impact of AI-based healthcare on patient autonomy (e.g., diagnostics systems, health apps, smart assistants; [62, 111, 112]). The following is a synthetic overview

¹⁶ The Reasonable Endorsement Test is formulated in this way: “An action is morally permissible only if it would be allowed by principles that each person subject to it could reasonably endorse” ([39], p. 52).

representative of the main topics in this sub-debate. Apart from identifying the key issues, the aim is to stress similarities and differences with the rest of the literature as well as with previous bioethical debates on patient autonomy. On the one hand, AI can provide more accurate information, predictions, and diagnoses for competent and/or authentic decisions [111, 113–115], as well as personalised assistance and health education [111, 113–115]. On the other hand, AI can produce biased diagnoses and inaccurate predictions or prevent access and understanding of relevant information due to opacity [63–65], cause information overload [111, 113–115], favour a new form of paternalism [66, 116] and pressure to use AI-based decisional aids [115]. Therefore, the integration of AI in healthcare can both overcome and strengthen many issues for patient autonomy usually discussed in bioethics. Also in this case, the development of Generative AI is raising further issues and opportunities. As an example, a recent and increasing debate is raised by the proposal to employ LLMs to develop a Personalized Preference Predictor [117] able to infer treatment's preferences of incapacitated patients for surrogate decision-making.¹⁷ In addition, relational accounts help identify further ways in which AI healthcare applications can undermine or support patient autonomy, through e.g., new forms of surveillance [107] (see 4.2.4). Moreover, since relational conditions (e.g., social support) are a fundamental aspect of autonomy in healthcare, they should inform the design and assessment of health recommender systems to promote patient's autonomy [31, 50]. Similarly, London et al., [112] stress the challenge for advanced smart assistance to understand ethical aspects of social interactions to support the autonomy of older users.

Another illustrative context taken into account in the literature is the workplace, where AI automation can reduce workers' autonomy by undermining the control over relevant work-related activities and choices (from the choice of applying to and accepting a job, to choices on how to perform one's job), as well as control over job applicant's self-representation. For instance, Vaassen [32] argues that, since algorithmic opacity prevents job applicants' from getting the causal explanation for their rejection, they cannot adjust their choices (e.g., what to change in the application) to pursue their goals and plans. As previously mentioned (Sect. 3.2.4), Aizenberg et al. [69] criticise algorithmic hiring assessment for hampering jobseekers' autonomy over self-representation by adopting standardised metrics that disregard people's histories. In a different way, Scripter [68] suggests that AI-based automation of the workplace reduces

workers' autonomy by diminishing control of work-related activities, as well as by imposing surveillance backed by sanctions (two aspects of what he calls an "autonomy gap"). Lastly, concerns arise also in the education context, especially due to the deployment of Generative AI. As reported in Sect. 4.2.6, Aylsworth and Castro [48] discuss the case of students using chatbots to write philosophical papers, which could undermine key capacities (e.g., critical thinking) underpinning their autonomy.

4.2.9 Theoretical challenges

The impact of AI on human autonomy also raises some crucial theoretical questions. A major issue is whether human-AI interaction poses inevitable, unique, and new threats to human autonomy. The literature doesn't provide a clear answer to that, but addresses the question from different angles. For instance, Prunkl [2] argues that there is no inherent trade-off between human and machine autonomy. As stressed by the author, this means that the risks usually come from external factors (e.g., misuse, malfunctioning) rather than from artificial systems acting autonomously. Moreover, Prunkl [3] proposes to distinguish between issues coming from the use (as a tool) or from the nature (as an agent) of AI. A rather critical and pessimistic view is defended by Lu [80] who believes that personalised algorithmic decision-making poses inevitable challenges to human autonomy: first, the value-laden nature of algorithmic recommendations and decisions steers choices toward external values rather than reflecting true preferences; second, the self-narrowing effect of personalised filtering creates a fixed identity, filter bubbles, and self-reinforcing loops; third, the issues of algorithmic dependence and deskilling. Against the predominant negative narrative surrounding AI assistance, Heinrichs [55] stresses that the risk of losing autonomy due to AI depends on a potential gain in autonomy itself: on the one hand, if we are not autonomous in the first place, we can't lose autonomy if we can't first gain it; on the other, if we already are autonomous, we wouldn't risk losing autonomy without the promise of increasing it. Similarly, Danaher [26] criticises and debunks the view that AI assistance necessarily undermines human autonomy, while recognising some unprecedented threats. More precisely, among the three conditions of autonomy proposed by Raz, recommender systems don't necessarily threaten rationality and optionality, but threaten independence due to manipulation [26, 27]. This brings us back to the question of the uniqueness and novelty of AI's impact on human autonomy. As summarised by Danaher [27, 52], the difference of AI's impact consists primarily in its scope, scale, and speed that make it ubiquitous, pervasive, and irresistible, as well as

¹⁷ In the original proposal, the Patient Preference Predictor is a computer-based algorithm inferring patient's treatment preferences (see e.g., [118, 119] for a summary of autonomy-based criticisms and replies).

in the new methods (e.g., personalisation and targetisation rather than standard manipulation).

Together with benefits and threats to the exercise of human autonomy, the interaction with AI also has crucial implications for the conceptualisation of autonomy itself. The development and deployment of (partially or fully) autonomous AI raises questions on how (human) autonomy is (and should be) defined. For instance, attributing autonomy to machines can affect the standard human-centred understanding of autonomy, while changing its core meaning (e.g., from self-governance to independence). Overall, the reviewed literature criticises the attribution of autonomy to artificial systems and agrees that human and artificial autonomy should not be conflated [120]. In this vein, Prunkl [2] shows that human and machine autonomy are distinct since the latter lacks the cognitive processes and requirements that characterise the former. Similarly, Soma et al. [121] recognise that machine autonomy is different from human autonomy since it lacks the capacity to reflect on, understand, and be responsive to reasons. Even more strongly, Chiodo's argument [24] against technological automation draws on and stresses the etymological difference between human autonomy (i.e., “self-given law”) and technological automation (i.e., being governed by something “offhand”). While acknowledging the need to rethink the concept of autonomy, Pierosara [47] proposes a distinction between human narrative autonomy (i.e., making meaning in an open, non-deterministic way) and artificial storytelling (i.e., collecting and transforming data into a story).

The difference between human and artificial autonomy is also linked to what Danaher [26] calls the “paradox of internal automaticity”: since automation also occurs within our bodies and minds, external and internal automaticity should have the same implications for autonomy. Danaher [26] also discusses some possible responses: accepting the problem, stressing the different locus of control (internal vs external), or adopting a “conservative bias” (biological vs artificial). The second strategy is followed by Bauer and Dubljević [51], who argue that the difference is the “internal locus of control”: humans have developed internal control over automatisms in a way that autonomous decision-making is partly constituted by internal automaticity, while AI outsourcing is more concerning because it has an external locus of control.

4.3 Discussion

The previous overview has shown that the impact of AI on human autonomy is ambivalent and multi-faceted, depending on a variety of elements: different AI-based technologies

can have different effects on different aspects of autonomy in different contexts. Therefore, the various competing analyses of AI's impact on human autonomy reflect and address the complexity and variety of issues at stake. However, the lack of a coordinated assessment of AI's impact risks undermining a full understanding of and adequate response to these impacts (see also [3]). Similarly, the literature doesn't extensively engage with previous philosophical debates and related literatures on the threats to autonomy. By contrast, this comparison would help identify underexplored issues in the literature (e.g., adaptive preferences, paternalism), but also new opportunities (e.g., the debate on Socratic nudges resembles the concept of boosts in behavioural economics), as well as explain what is new, specific, or unique about AI's impact on human autonomy.

To provide a common framework, the various issues discussed in the literature have been classified under two main categories: the impact on deliberation, decision-making, and behaviour when AI mediates human activities, and the impact on performing choices, actions, and goals when AI automates human activities. While the literature recognises both positive and negative impacts of AI on human autonomy, it tends to focus on the latter. A major problem is the influence of AI decisional systems on the individual capacity to make autonomous decisions, affecting both personal and moral autonomy. However, there is disagreement on how to define and assess AI influence, especially in the case of manipulation and nudging.

At the same time, the literature pays less but increasing attention to the impact of AI on other components, dimensions, and conditions of autonomy. More precisely, relational and some multi-dimensional accounts show how the development and deployment of AI interact with the socio-relational conditions and “sociotechnical bases” of autonomy. Apart from giving a more comprehensive assessment, these multiple perspectives could also help counter the tendency to focus on the negative effects of AI on human autonomy. Moreover, they can provide new guidelines for the design and development of AI that respect human autonomy.

The proposed classification also took into account the impact of specific technologies and the context of analysis. Illustrative cases are recommender systems as well as other AI-based decisional assistants. A recent turning point to be further explored is the development of Generative AI and its applications, such as digital duplicates and AI agents, that exacerbate the issues of previous digital assistants acting on one's behalf. Moreover, the advancement of AI-based technologies (e.g., neurotechnologies) could challenge the internal-external difference and complicate the relation between autonomy, automaticity, and automation due to

the unprecedented integration and mediation of cognitive processes.

Together with different AI technologies, the literature considers various domains, such as entertainment, health-care, and workplace. The concerns for online environments like e-commerce and social media platforms are in line with debates on the socio-political implications of cases like Facebook emotional contagion experiment, the Cambridge Analytica scandal, and related phenomena such as filter bubbles and echo-chambers (although the latter have been recently questioned; see e.g., [122]). Less explored among these domains is the impact on students and teachers' autonomy, although this is likely to change precisely due to the deployment of Generative AI in the education sector.

Intertwined with the ethical challenges, the review has also identified some theoretical issues. Regarding the novelty of AI's impact, AI can reproduce, modify, and amplify traditional threats to autonomy (e.g., manipulation, deception, surveillance), but also overcome others (e.g., internal constraints such as cognitive biases, external constraints such as information overload). Moreover, AI expands the scope and effects of technological automation to cognitive tasks and capacities (e.g., decision-making and creativity). In this sense, the autonomy of artificial systems might play a role due to the capacity to completely replace and bypass human autonomous decision and action. For all these reasons, it can be argued that AI doesn't simply reproduce old phenomena but enables new versions of them (e.g., "micro-targeting", "hypernudging", "algorithmic predeterminism", "hypersuasion") and leads to unprecedented effects. Nevertheless, since many of these issues are more of an evolution of already existing phenomena, rather than radically new ones, the substantial difference and novelty of AI's impact remains an open question. To clarify this point, the literature would benefit from a greater engagement with previous philosophical debates discussing major threats to autonomy (e.g., manipulation, coercion, paternalism, weakness of the will).

Alongside the impact on the exercise of human autonomy, further questions concern the relation between human and machine autonomy, as well as the conceptual implications for the notions of autonomy itself. New distinction, like the one between narrative autonomy and artificial storytelling, can turn useful for this purpose. Therefore, a future theme precisely consists in rethinking the concept of human autonomy in relation to machine autonomy and in response to the challenges raised by AI for its definition, value, and exercise.

The issues discussed in this Section do not exhaust nor respond to all the questions concerning the relation

between AI and human autonomy, but show the need for further investigation. Therefore, more philosophical work is required to analyse how AI affects not only the exercise but also the concept and value of human autonomy.

5 Contributions and future directions

Within AI ethics, there is an increasing academic literature concerning the impact of AI on human autonomy. However, it lacks a common framework providing a comprehensive understanding of the issues at stake. This literature review offers the following contributions to this debate.

First, this overview summarises the state of the art by mapping and unpacking the key concepts, themes, and issues. The first part has identified and spelt out different conceptualisations of human autonomy used for analysing AI's impact. The second part has mapped the ways in which AI can affect the exercise and concept of human autonomy. Second, this overview also serves as a common framework that structures the literature by synthesising various accounts and analyses. This should work as a compass for navigating the complex and multifaceted relation between AI and human autonomy. Third, the review compares and connects the literature with previous philosophical debates on autonomy to clarify the current discussion. For instance, it shows that the understanding of human autonomy in AI ethics is influenced by contemporary accounts of personal autonomy. It also shows that AI can reproduce, amplify, or modify several traditional threats to autonomy (e.g., manipulation, coercion, surveillance), but also overcome others (e.g., cognitive limits, pathologies, and impairments). Lastly, the analysis of the literature reveals some issues, gaps, and open questions (see Table 4). Among these challenges, the review stresses the need to rethink human autonomy in the face of AI's impact through more comprehensive (e.g., multidimensional) accounts. This should inform the assessment of AI's impact on its different components and conditions, as well as guide the design and use of AI to support rather than undermine human autonomy.

Appendices

Appendix 1: Methodology

This paper consisted in a review of the AI ethics academic literature on the relation between AI and human autonomy. This literature review followed the methodological

approach of critical or narrative philosophical reviews. The aim was to provide a synthesis, an analysis, and an evaluation of the reviewed literature to understand and advance the ongoing debate, emphasising the arguments, theories, and concepts presented. The identification of relevant contributions started by including literature I knew to be suitable for this purpose. For a more complete overview, I also conducted a more structured search with inclusion and exclusion criteria that reflect the topic of the review (the debate on human autonomy in AI ethics). For this reason, I searched for scholarly literature within AI ethics that addressed the relation between AI and human autonomy. The initial search used the academic philosophy database PhilPapers with “autonomy+AI” as the query, and returned 224 results. The choice of the database is motivated by the aim of identifying the philosophical literature addressing AI’s impact on human autonomy within the broader field of AI ethics. To select relevant literature, I adopted the following inclusion criteria: addressing AI’s impact on human autonomy among the key topics of the paper; referring to some notion of human autonomy, either based on a philosophical theory or on a basic understanding; engaging in an ethical analysis of AI’s impact; specifically discussing AI technologies. These inclusion criteria aimed at identifying relevant literature providing both a conceptualisation of human autonomy and an ethical discussion on how it is affected by AI. By contrast, the review adopted the following exclusion criteria: focusing on machine autonomy; not sufficiently discussing the impact on human autonomy; not specifically referring to AI systems but to broader technologies (e.g., digital technologies, big data); not engaging with philosophical or ethical analysis (e.g., only epistemic discussions, policy issues, empirical studies on consumer decisions). Together, these inclusion and exclusion criteria aimed at circumscribing the search to academic literature within the field of AI ethics alone, and at filtering the literature to match the topic of the review as much as possible. For a first selection, the records were scanned on the basis of the title and abstract according to previous criteria: a group of 41 records was identified in this way. Several results were discarded because they focused on the autonomy of AI ($N=42$) or because they didn’t meet at least one of the inclusion criteria (see Fig. 1 for more details). Duplicates, citations, non-English records, and non-accessible content were also removed at this stage. The selected papers were then read in full to double-check their alignment with the inclusion criteria and with the scope of the review. In this way, 30 results were selected, while 10 results were discarded because they didn’t have a strong ethical focus on autonomy ($N=10$) or didn’t explicitly focus on AI ($N=2$). This initial search was expanded by looking for related literature

through reference lists, citation functions (e.g., “cited by”, “related papers”), or similar queries (e.g., “human autonomy” AND “artificial intelligence”, “human autonomy” AND “AI”), also in other databases (e.g., Google Scholar, Semantic Scholar). This process was reiterated during the writing phase to ensure that no recent important publication (until December 2024) was missing. All rounds of selections followed the initial inclusion and exclusion criteria. In this way, an additional group of 31 records was identified and included in the review. Lastly, the dataset resulting from the search was compared and integrated with the initial group of previously known texts: among them, those ($N=9$) that had not appeared in the search (e.g., because focused on more specific AI applications) were added to the dataset. A total of 70 records formed the database of the review (See Table 1 for a complete list). The reviewed literature includes both published and unpublished material (e.g., academic papers, conference proceedings, dissertations, books), consisting in both primary (e.g., original research) and secondary literature (e.g., reviews). The review covers literature until December 2024, also including papers that were first published online in 2024, and then published in an issue in 2025. References from related fields (e.g., moral philosophy, bioethics, human–computer interaction, etc.) cited in this paper and included in the bibliography were used in the analysis of the results, but are not themselves part of the review. The selection of themes followed from an in-depth analysis of the texts aimed at identifying key concepts and issues, common patterns and trends, as well as gaps and open questions. For this purpose, I took into account several factors, from keywords in titles and (sub)headings (e.g., “Descriptive Autonomy”, “dimension(s)”, “relational”, “manipulation”, “nudging/nudges”, “heteronomy”), to the general topic, context, or debate addressed in the paper (e.g., AI-based healthcare, ethical decision-making). In so doing, I started the theme selection by drawing on pre-existing themes in previous philosophical literature on autonomy, to identify similar conceptions (e.g., personal autonomy, relational accounts) and issues (e.g., manipulation, nudging) in the reviewed literature. This approach remained flexible enough to leave room for variations and new conceptions or issues emerging from the literature (e.g., ethical and narrative autonomy, new multidimensional accounts, cognitive outsourcing, etc.). The combination of pre-existing and new terminology allowed me to provide a more comprehensive overview of the current literature. At the same time, I relied on the comparison with the previous philosophical literature as the background and guideline for explaining and analysing the themes identified by the review. The analysis of results adopted a qualitative approach (in the Findings), combined with a critical analysis (in the Discussion).

Fig. 1 Chart of the methodology used for the identification of relevant literature, including searched databases, inclusion/exclusion criteria, and number of records

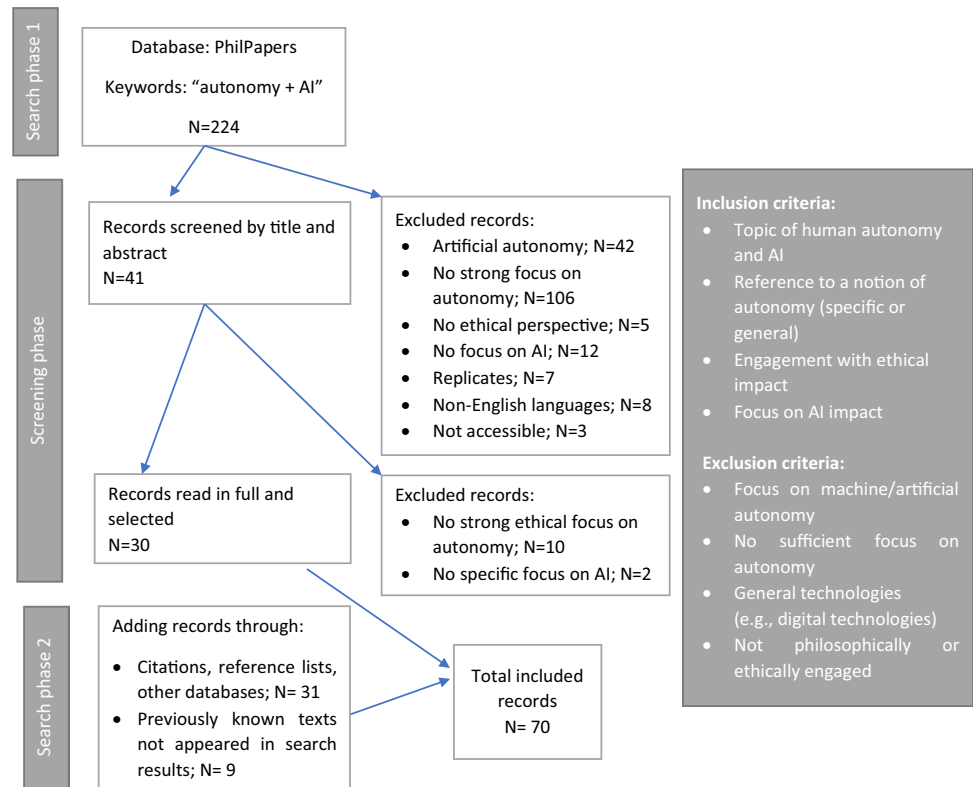


Table 1 Database of literature included in the review

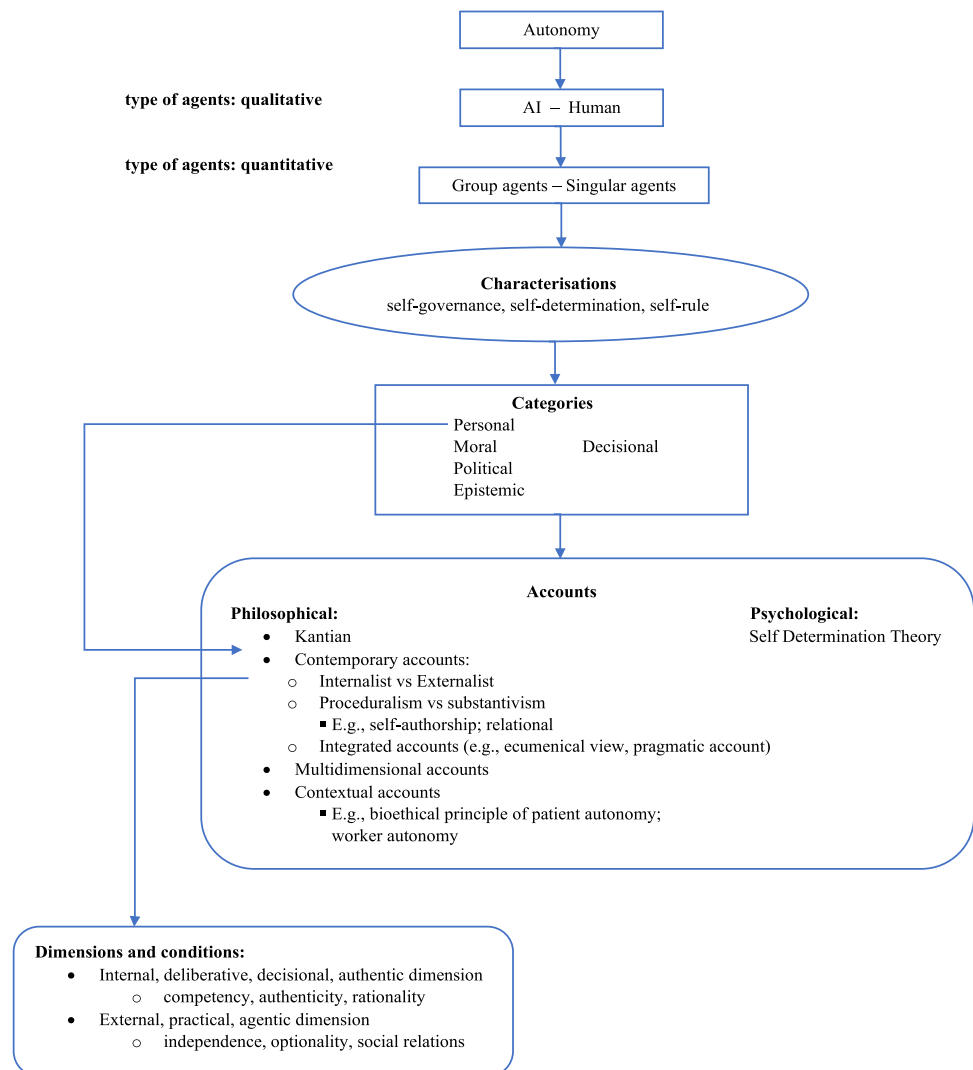
References	Source
Aizenberg et al. [69]	PhilPapers
Allen et al. [113]	PhilPapers
Aylsworth and Castro, [48]	Other database/Reference list/Citation
Bartmann, [88]	Other database/Reference list/Citation
Bauer and Dubljević, [51]	PhilPapers
Benzinger et al. [111]	PhilPapers
Bermúdez et al., [82]	PhilPapers
Bonicalzi et al. [25]	Previously known
Calvo et al. [58]	PhilPapers
Chiodo, [24]	PhilPapers
Chiodo, [45]	Other database/Reference list/Citation
Danaher, [26]	PhilPapers
Danaher, [27]	Other database/Reference list/Citation
Danaher, [52]	Other database/Reference list/Citation
Danaher and Nyholm, [86]	Other database/Reference list/Citation
del Valle et al. [28]	PhilPapers
del Valle and Lara, [53]	PhilPapers
Earp et al. [117]	Other database/Reference list/Citation
Ernst, [104]	Other database/Reference list/Citation
Floridi, [87]	Other database/Reference list/Citation
Formosa, [29]	PhilPapers
Fossa, [108]	Other database/Reference list/Citation
Fossa, [8]	Other database/Reference list/Citation
Giubilini and Savulescu, [34]	Previously known

Table 1 (continued)

References	Source
Guerrero Quiñones, [114]	PhilPapers (second round)
Harris, [40]	PhilPapers
Heinrichs, [55]	Other database/Reference list/Citation
Ho, [107]	Other database/Reference list/Citation
Ienca, [11]	PhilPapers
Kempt et al. [83]	Other database/Reference list/Citation
Keskinbora, [62]	Other database/Reference list/Citation
Klincewicz and Frank, [63]	PhilPapers
Klugman, [64]	PhilPapers
Koeszegi, [81]	Other database/Reference list/Citation
Kühler, [116]	PhilPapers
Laacke et al. [65]	PhilPapers
Lanzing, [98]	Other database/Reference list/Citation
Laitinen and Sahlgren, [23]	PhilPapers
Lara, [35]	PhilPapers
Lara and Deckers, [36]	Other database/Reference list/Citation
Lara and Rodríguez-López, [84]	PhilPapers
Lindemann, [85]	PhilPapers
London et al. [112]	PhilPapers
Lu, [80]	Other database/Reference list/Citation
Magrani and da Silva, [30]	Other database/Reference list/Citation
McDougall, [66]	PhilPapers
Mhlambi and Tiribelli, [7]	PhilPapers
Milano et al. [79]	Previously known
Mittelstadt et al. [41]	Previously known
Pierosara, [47]	PhilPapers
Poszler and Lange, [46]	Other database/Reference list/Citation
Prunkl, [1]	PhilPapers
Prunkl, [2]	PhilPapers
Prunkl, [3]	Other database/Reference list/Citation
Rini, [49] (unpublished manuscript)	PhilPapers
Riva and Tiribelli, [37]	Previously known
Rubel et al. [39]	Previously known
Santoni de Sio and Fossa, [56]	Other database/Reference list/Citation
Sass, [42]	Philpapers (second round)
Sass, [115]	Other database/Reference list/Citation
Savolainen and Ruckenstein, [54]	Other database/Reference list/Citation
Scripter, [68]	PhilPapers
Shevlin, [60]	Other database/Reference list/Citation
Soma et al. [121]	Other database/Reference list/Citation
Subías-Beltrán et al. [105]	Other database/Reference list/Citation
Tiribelli, [38] (doctoral dissertation)	Previously known
Tiribelli, [50]	Previously known
Tiribelli and Calvaresi, [31]	Other database/Reference list/Citation
Vaassen, [32]	PhilPapers
Varshney, [33] (unpublished manuscript)	Previously known

Appendix 2: Results

Fig. 2 Framework of conceptions of autonomy



This figure graphically represents the framework proposed by the review to map and spell out the relation between different elements of the concept of autonomy in AI ethics. Preliminary distinctions regard the type of autonomous agents, from a qualitative (humans or artificial systems) and quantitative (single or group) perspective. In AI ethics, autonomy mostly refers to individual agents, either humans or AI systems. Human autonomy is broadly characterised as self-governance, self-determination, self-legislation, self-rule. These basic characterisations are further specified in different conceptions (i.e., ways of thinking about a concept to specify what it is about), that are variations of (individual) self-governance/self-determination. These conceptions can refer to the main categories (i.e., classes of conception pertaining to the same domain) of autonomy discussed in previous philosophical literature: personal or individual, moral, political, and eventually epistemic. In line with

contemporary philosophical debates on autonomy (since the twentieth century), the reviewed literature focuses on personal/individual autonomy. Additionally, decisional autonomy can be considered as either a super-category (i.e., a broader meta-category) or a sub-category (i.e., a component or dimension) of other categories of autonomy. These categories, especially personal autonomy, are articulated in accounts that provide a definition and spell out components and conditions of autonomy. The reviewed literature presents several accounts spelling out the categories of personal and moral autonomy, mostly but not exclusively derived from the previous philosophical literature. The following main groups are identified: Kantian, Contemporary, Multidimensional, Contextual, Psychological. Following contemporary accounts, the literature recognises multiple different components, dimensions, and related conditions of autonomy, including internal and external elements

Table 2 Overview of the main conceptions of autonomy in the reviewed literature, including the respective categories and accounts, as well as the themes or issues addressed

	Conceptions of autonomy	Issues and themes raised by AI
Moral and ethical autonomy	(Chiodo, [24], [45]) Kantian self-given rule, i.e., reflecting on and choosing in accordance to reasons, being responsible and accountable	Technological automation and “overturning” human autonomy
	(Fossa, [8]) capacity to make autonomous moral decisions on e.g., harm allocation	AVs change the scope of moral autonomy
	(Giubilini and Savulescu, [34]; Lara and Deckers, [36]; Lara, [35]; Lara and Rodríguez-López, [84]; Poszler and Lange, [46]) ability to reflect on and decide according to one’s moral perspective and ethical principles	AMAs for moral enhancement
	(Tiribelli, [38]; Riva and Tiribelli, [37]) being the authors of our decisions and actions, being able to reflectively endorse the motives of our moral choices	Influence of recommender systems
Personal autonomy	(Pierosara, [47], p.1786) “the capacity to deliberate and act without constraints following an interpretation and endorsement of what we consider valuable”. It’s a narrative autonomy, i.e. human capacity to make meaning and to create stories in an open and non-deterministic way	Difference between human and artificial autonomy
	(Aylsworth and Castro, [48], p. 117) Kantian “humanity”, i.e., “the rational agent’s capacity to set and pursue her own ends”	Philosophy students should not outsource the writing of philosophical papers due to the duty to respect and promote autonomy
	(Rini, [49], unpublished) Kantian-Piagetian account of autonomy as self-control (i.e., capacity of deliberation and accountability to reason)	Authoritarianism with chatbots
	(Bauer and Dubljević, [51]; Danaher, [26], [27], [52]; Fossa, [8]) Raz’ <i>self-authorship</i> : rationality (i.e., understanding the reasons for choice), optionality (i.e., adequate range of available and viable alternatives), independence (i.e., freedom from coercion and manipulation)	Issues and benefits of cognitive outsourcing
	(Ho, [107]; Kempt et al. [83]; Mahmbi and Tiribelli, [7]; Pierosara, [47]; Tiribelli, [50]; Tiribelli and Calvaresi, [31]; Soma et al. [121]) Western-Feminist and non-Western relational autonomy, i.e., caused or constituted by relational conditions such as social support, social recognition, reciprocity, etc. (e.g., in Ubuntu, autonomy is grounded in interconnectedness, relatedness, reciprocity, and active participation, and implies duties toward others)	Impact of algorithmic relations and socio-technical relations on relational conditions of autonomy, including marginalised communities
	(Bonicalzi et al. [25]) <i>Descriptive Autonomy</i> , i.e., range of internal and external features (e.g., competency, authenticity, social relations); distinct from <i>Normative Autonomy</i> , i.e., ethical principle to respect	Ethical issues of recommender systems on decisions, identity, and critical thinking
	(Formosa, [29]) proceduralist, substantivist, and externalist elements: a range of competencies, authentic values, and sufficient available options	Social robots enable or disable autonomy through more or less valuable ends, authentic choices, and competencies
	(Rubel et al. [39]) <i>ecumenical view</i> reconciling procedural-psychological and socio-relational accounts of personal autonomy, including both procedural independence (epistemic competence and authenticity) and substantive independence (reciprocal support and non-domination)	Impact of automated decisions of algorithms that do not pass Reasonable Endorsement Test
	(Del Valle and Lara, [53]) <i>pragmatic account</i> incorporating relational elements into a revised procedural autonomy: competency (i.e., information management and deliberative skills) and authenticity (i.e., reflection of personal identity)	Issues of Recommender Systems: manipulation and cognitive dependence concern competency; challenges to reflective endorsement, authenticity, and explainability concern authenticity
	(Prunkl, [3]) multidimensional framework: authenticity or the internal dimension (i.e., capacity to decide and act on the basis of one’s own motives) and agency or the external dimension (i.e., effective capacity of making choices and enacting them)	AI manipulation affects authenticity; lack of control of AI affects agency
	(Tiribelli and Calvaresi, [31]) liberal and relational autonomy including three dimensions and corresponding conditions: procedural independence requires competency, substantive independence requires authenticity, relatedness requires socio-relational elements	Autonomy-based ethical analysis of Health Recommender Systems
	(Heinrichs, [55]) “epistemic” (i.e., <i>self-knowledge</i>) and “practical” (i.e., <i>self-perfection</i>); “active” (<i>competency</i>) and “passive” (<i>authenticity</i>)	Impact of Digital assistants: over-emphasis on “passive” component overlooks potential gains for “active” component
	(Sass, [42]) <i>Tri-partite model</i> : agency (i.e., capacity to take action and develop skills), authenticity (i.e., the result of a process in which the individual has several preferences, a developed life plan, a set of capacities for critical reflection that can exercise them to select preferences), and control over decision-making (i.e., awareness of available choices and approval of external influences)	Manipulation can diminish some dimensions of autonomy while improving others

Table 2 (continued)

	Conceptions of autonomy	Issues and themes raised by AI
	(Laitinen and Sahlgren, [23]) “maximally informative” account including capacities for self-determination, normative requirements, respect and self-respect, performativity or exercise, and so-called “sociotechnical bases”	Various themes such as cognitive heteronomy, respect of autonomy, socio-technical impact
	(Santoni de Sio and Fossa, [56]) Autonomy in driving: “self-determination of driving decisions”, “freedom to pursue a good life through mobility”, “capacity and opportunity to influence social deliberation concerning transportation policies and planning”	Driving automation can reduce driving self-determination but improve freedom
	(Savolainen and Ruckenstein, [54]) Four dimensions of autonomy in algorithmic relations: two pertaining to an <i>instrumental horizon</i> (i.e., algorithms are technical procedures for ordering social life) and consisting in <i>algorithmic capacity</i> (i.e., capacity to identify and understand algorithms) and <i>situational mastery</i> (i.e., using algorithmic affordances for one’s goals); two pertaining to an <i>intimate horizon</i> (i.e., algorithms give rise to affective relationships that involve sharing a personal space) and consisting in <i>breathing space</i> (i.e., room for realisation and reflexivity) and <i>co-evolution</i> (i.e., self-understanding through algorithms)	Reconfiguring a relational and multidimensional autonomy to address issues of algorithmic relations
	(Allen et al. [113]; Guerrero Quiñones, [114]; Klinecicz and Frank, [63]; Klugman, [64]; Kühler, [116]; Laacke et al. [65]; London et al. [112]; McDougall, [66]; Sass, [115]) Bioethical principle of patient autonomy: capacity and right to make medical decisions aligned with one’s values and goals, thanks to intentionality, understanding, and independence	Various themes such as influence on decision-making, preference prediction, biases, opacity, paternalism
	(Calvo et al. [58]; Varshney, [33]; Shevlin, [60]) autonomy in <i>Self-Determination Theory</i> : self-endorsement of motives and behaviours, requiring feeling of willingness, volition, endorsement; lack of pressure and control; lack of deception and misinformation	Various themes such as coercion, manipulation, addiction, etc. in six spheres of technological experience (adoption, interface, tasks, behaviour, life)
Decisional autonomy	(Mittelstadt et al. [41]) Decisional autonomy [no explicit definition]	The paradoxical effects of algorithmic influences: personalization should improve decision-making with more relevant information, but reduces information diversity and nudges toward external preferences
	(Floridi et al. [43]) <i>meta-autonomy</i> , i.e., “power to decide (whether to decide)”, balance between freedom of choice and (reversible) delegation to AI	Keeping or losing control of AI technologies and related decisions

Table 3 Overview of AI’s impact on different aspects of autonomy

Source of impact	Negative impact	Positive impact	Component, dimension, condition of autonomy	List of references (alphabetic order)
AI mediation	Interference with rational deliberation, decision, behaviour • manipulation, coercion, deception • nudging, assistance • prediction, filtering, recommendations • surveillance	Intervention in support of decision-making • nudging, assistance, education	deliberative and decisional capacity; competency, authenticity, independence, control	Bermúdez et al. [82]; Bonicalzi et al. [25]; Danaher, [26], [27]; Ernst, [104]; Fossa, [108], [8]; Giubilini and Savulescu, [34]; Koeszegi, [81]; Laitinen and Sahlgren, [23]; Lara and Deckers, [36]; Lara, [35]; Lara and Rodríguez-Lopez, [84]; Magrani and da Silva, [30]; Mittelstadt et al. [41]; Milano et al. [79]; Poszler and Lange, [46] Prunkl, [2]; Santoni de Sio and Fossa, [56]; Savolainen and Ruckenstein, [54]; Subías-Beltrán et al. [105]; Tiribelli, [38]; Tiribelli, [50]; Tiribelli and Calvaresi, [31], Vaassen, [32]; Varshney, [33]
AI automation	Replacing and bypassing decisions and actions • undermining decisions and goals • deskilling and dependence • heteronomy	Supporting realisation of decisions and actions • saving resources • increasing options	decisional capacity, practical and agentic capacity; authenticity, competency, independence, optionality, agency	Aizenberg et al., [69]; Aylsworth and Castro, [48]; Chiodo, [24], [45]; Danaher, [26], [27]; Formosa, [29]; Fossa, [108], [8]; Heimrichs, [55]; Laitinen and Sahlgren, [23]; Lindemann, [85]; Prunkl, [2, 3]; Rubel et al., [39]; Santoni de Sio and Fossa, [56]; Scriptor, [68]

Table 4 Overview of gaps, open questions, and future directions

Category	Gaps and open questions	Future directions
Concept of autonomy	Re-definition of human autonomy: • other accounts than standard philosophical individualistic ones • comparison with machine autonomy • Operationalisation of human autonomy	Engaging more with other philosophical accounts of autonomy, e.g. feminist and non-Western relational autonomy, multidimensional accounts • Engaging more with psychological accounts of autonomy • Broadening the concept to include both human and machine autonomy vs stressing the differences against conflation
The impact of AI	Overlooked or new effects: • socio-relational impact • outsourcing and deskilling • effects of new technologies • impact on other notions and conditions • positive impact (opportunities, benefits)	Researching on: • AI deskilling • neurotechnologies, generative models, General AI • privacy issues, socio-relational effects • ways AI can support and improve human autonomy

Appendix 3: glossary

Accounts: the term is used by the review to indicate the theoretical framework, including theories and models, that specify the categories of autonomy by spelling out definition, components, dimensions, and conditions.

Algorithmic predeterminism: According to Tiribelli [38], algorithms lead to a new form of “predeterminism” in the sense that what the user choses is either already known by the algorithm or influenced by it in a way that pushes the user to align with the predicted behaviour.

Agency: Depending on the account, it is one of the conditions or dimensions of autonomy consisting in the capacity to act on the basis of one’s beliefs, values, and goals.

Algorithmic outsourcing: The process of offloading (cognitive) tasks to smart algorithms [26].

Artificial Intelligence (AI): There isn’t a standard definition of Artificial Intelligence (AI), but the term has been used in different ways and to describe various technologies (see also [9, 10]). Broadly, AI technologies are computer systems that perform tasks resembling intelligent behaviour, with some degrees of independence, and in a goal-directed way. Many AI technologies are based on machine learning, a method through which algorithms learn from data—via supervised, unsupervised, or reinforcement learning—to make inferences on new data without being explicitly programmed. In this review, the term “AI” or “AI technologies” is used in a broad way that mostly concerns, without being limited to, machine learning systems. A wide range of

AI applications are included, from recommender systems, to health apps, to social robots, to chatbots.

Artificial Moral Assistants/Advisors (AMAs): AI systems designed to help with ethical decision-making to enhance the capacity to decide in line with one’s moral values, principles, and goals [46]. The Artificial Moral Advisors by Giubilini and Savulescu [34] provides advice in the forms of recommendations to always act in accordance with one’s moral goals. The “SocrAI” by Lara and Deckers [36] is a conversational bot for the moral education of the user: it doesn’t provide advice but guidance to improve users’ decisional capacity by teaching them how to better reasoning about moral issues in an independent way.

Authenticity: Depending on the account, it is one of the conditions or dimensions of autonomy consisting in an internal alignment with one’s motives.

Autonomous Vehicles: Also called self-driving cars, these are vehicles that can operate with minimal or no human intervention, depending on the level (from 0 to 6) of automation.

Characterisations: This expression is introduced by the review to indicate general basic descriptions of a concept. In the case of human autonomy, some common characterisations are self-governance, self-legislation, self-rule, self-determination.

Competency: One of the internal conditions of autonomy, consisting in a range of capacities held by the autonomous agent (for an overview see [16]), including rational thought, self-control, and independence – as well as the absence of e.g., mental impairment, self-deception, addictive attitudes, etc.— but also self-attitudes (e.g., self-trust and self-esteem).

Components: This expression is used to describe some aspects that are a constituent part of autonomy. For instance, decisional autonomy can be considered as a component, super- or sub-category, overlapping with different broader categories of autonomy (personal, moral, political). In the review, this term is also used as a synonym of dimensions (see below).

Conceptions: This expression indicates different ways of thinking about a concept to specify what it is about (see [44]). Therefore, different conceptions of autonomy are different ways of understanding its basic characterisations of self-governance, self-rule, and self-determination. The review has distinguished different groups of conceptions on the basis of different categories and the related accounts of autonomy.

Conditions: This term is used to indicate some requirements to fulfil in order to be and act autonomously (see e.g., [25]).

Digital duplicates, deathbots, or griefbots: Digital duplicates are a broad category of AI systems consisting in “partial, at least *semi-autonomous*, *digital recreations* of real

people” [86]. Deathbots or griefbots are a sub-set of digital duplicates that recreate a dead person [85].

Dimensions: The term “dimensions” comes from Mackenzie’s account of personal autonomy [57] and is now relatively employed in both philosophical and AI ethics’ literature on autonomy. However, despite being an established term, there isn’t a standard definition specifying and listing these dimensions. Depending on the account of autonomy, its dimensions can include constitutive aspects of autonomy, forms that autonomy can take, as well as elements that are among the preconditions of autonomy without being constitutive of it.

Decisional autonomy: autonomous decision-making is the ability to decide for oneself, without external undue interference, and based on motives/goals that one would endorse.

Ethical autonomy: “The capacity to deliberate and act without constraints following an interpretation and endorsement of what we consider valuable” [47].

Epistemic autonomy: Broadly, the capacity to think for oneself (i.e., forming and revising beliefs), independently from undue external influence.

Generative AI/generative models: A class of AI systems that generate new content (e.g., textual, visual, audio, etc.), that resembles the original one usually based on the user’s prompt. More precisely, a generative model uses machine learning (mainly deep learning) algorithms that are trained on a huge amount of data to learn the probabilistic distribution that is then used to generate new data, which is similar to the input (it could have been part of the training set) but not identical (thanks to a stochastic component).

Independence: A condition of autonomy that expresses the “freedom from” obstacles and constraints, such as pressure, coercion, manipulation, or controlling influences.

Machine/artificial autonomy (footnote1): the autonomy of machines or of artificial systems. It is the property of a machine to perform some tasks independently, to a certain extent, from human intervention, control, or oversight (see e.g., [23]).

Manipulation: The concept has been defined in different ways, but it broadly refers to an undue form of influence that bypasses rational deliberation and compromises the authenticity of choices. Ienca [11] provides a narrative literature review of the concept in philosophy, psychology, and AI research. This allows to identify some common characteristics: Intentionality, outcome asymmetry, covertness and lack of transparency, violation of autonomy. In relation to AI, Ienca [11, p.840] also defines “digital manipulation” as “any influence exerted through the use of digital technology that is intentionally designed to bypass reason and to produce an asymmetry of outcome between the data processor (or a third party that benefits thereof) and the data subject”.

Moral autonomy: In Kant’s theory, it is the capacity of the will to give a universal moral law to oneself. However, the reviewed literature uses this concept to indicate the ability to form, reflect, decide, and act in accordance with one’s moral values or ethical principles [46], or the condition of being the authors of one’s own decisions and actions by reflectively endorsing the motives underpinning moral choices [38].

Moral deskilling: The expression is introduced by Vallor [110] in analogy with other forms of deskilling that result from automation. It indicates the weakening or loss of moral skills and virtues underpinning moral decision-making: indeed, delegating moral decisions to autonomous systems (e.g., autonomous weapons systems) can cause an atrophy of moral capacities that are no longer exercised.

Narrative autonomy: The expression is used by Piero-sara [47] to indicate the capacity of making meaning and creating stories in an open and non-deterministic way. It is opposed to “artificial storytelling”, which is the algorithmic process of collecting data and turning them into a story.

Normative and Descriptive Autonomy: The distinction is introduced by Bonicalzi et al. [25] to distinguish two “general understanding” or “forms” of autonomy. The first indicates the moral principle of being treated as an end or as an agent able to make decisions based on one’s interests/preferences. The second indicates some conditions to fulfil in order to be and act autonomously.

Notion: General idea and understanding of a concept not further specified.

Nudging: The concept was first used by Thaler and Sunstein [94] to define techniques that influence behaviour by shaping the “cognitive architecture” and by targeting human biases or heuristics to steer better choices. As the name suggests, “digital nudges” are the digital implementation of the physical ones, even though some of them can have a more adaptive capacity [92]. The concept of “hypernudge” is introduced by Yeung [97] to indicate the technique of producing analytic nudges that are “continuously updated, dynamic and pervasive”.

Optionality: This expression is derived from Raz [20] to indicate the availability of meaningful and viable alternatives, including both material options as well as moral and political views.

Personal/Individual autonomy: “the capacity to be one’s own person, to live one’s life according to reasons and motives that are taken as one’s own and not the product of manipulative or distorting external forces, to be in this way independent” [16].

Political autonomy: In a broad sense, it is the capacity and right of self-determination in the public sphere. In political philosophy, autonomy is at the basis of political legitimization, of recognition of citizens’ interests and rights, and of the claim against oppressive social conditions [16].

Privacy: There can be different definitions and interpretations of privacy. The review has referred to the concepts of informational and decisional privacy as defined by Lanzing [98]: the first is the control over one's personal information, including whom and to what extent has access to it; the second is the control over others' intervention/interference with how we lead our lives.

Recommender systems: Algorithms that make personalised and targeted recommendations on the basis of users' preferences and behaviours. Even though there are different filtering techniques (e.g., content-based, collaborative, hybrid), the common task of recommender systems is the following: collecting user's behavioural data to create a profile used to make predictions on one's preferences and to recommend the most likely option. For example, social media recommender systems rely on users' previous interaction (e.g., "like", "share", "comment") with the platform to maximize future engagement by predicting and suggesting content that matches one's preferences.

Relational autonomy: Despite differences between the accounts, relational autonomy is conceived as dependent on socio-relational conditions (either in a causal or constitutive way) that can sustain or hinder it. Historically, relational accounts developed in response to the traditional individualistic notions of autonomy by stressing the role of interdependency and of external influences and conditions that shape the autonomous self. One of the main concerns of relational accounts is the relation between autonomy and oppressive socialisation. In this respect, proceduralist relational accounts, stress the role of internal self-reflection and self-endorsement, while substantive relational accounts stress the importance of having the right content of motives (e.g., non-oppressive preferences). Relational accounts can be aligned with non-Western conceptions of personhood and autonomy, as in the case of Ubuntu philosophy.

Self-Determination Theory: Psychological theory of motivation and behaviour developed by Ryan and Deci [59]. It recognises three "needs" (i.e., the links between external affordances or demands and people's tendencies toward wellbeing) for individual wellbeing: competence, autonomy and relatedness. In this view, autonomy means self-regulation of behaviour, is expressed in acting with full volition and endorsement, and is strictly related with intrinsic motivation. In the reconstruction by Calvo et al. [58], autonomy consists in the self-endorsement of a behaviour, in a feeling of control, and in a sense of volition.

Transformative effects: the expression is introduced by Mittelstadt et al. [41] to indicate the outcomes of the process through which algorithms "reontologise the world by understanding and conceptualising it in new, unexpected ways".

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Data availability The list of literature included in this review is provided in the Appendix of this manuscript.

Declarations

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